



DRIVEN PILE FOUNDATIONS

PURPOSE

This document establishes guidelines and recommended procedures for the design of driven pile foundations subjected to static loading.

SCOPE

This document consists of a narrative, attachments, and 8 sample designs. The narrative covers general pile information, pile capacities, and basic design methods for piles, pile caps, and some connection details. There are attachments showing recommended types of uplift connectors for concrete, steel, and timber piles. There are sample designs for the following:

- Vertical vessel foundation
- Horizontal equipment foundations (using a grade beam)
- Piperack braced bay foundation (using battered piles)
- Combined foundation
- Three pile foundation for T-support
- Sphere foundation
- Cylindrical heater foundation

In addition, there is a sample design showing the checking of the structural capacity of a pile.

APPLICATION

This document applies to the design of driven pile foundations.

GENERAL

Piles are generally used to transmit foundation loads to deeper soil mass, to reduce settlement due to consolidation of upper soil strata, to anchor structures subjected to large uplift loads, and to resist large horizontal forces.

Pile Types

Piles are usually constructed of timber, steel, concrete, or combinations of these materials. Selection of the type of pile to be used should be based on cost, availability of materials, job restrictions, client requirements, and the recommendations of the geotechnical consultant. Timber and concrete piles are usually more economical than steel piles, but steel piles generally have higher load carrying capacities and are easier to connect. The Design Engineer will be given the types, capacities, and details of the piles to be used by the Lead Structural Engineer. Normally, pile details will be presented on standard drawings for a project. Properties of steel H-piles can be found in the AISC (American Institute for Steel Construction) Manual.



DRIVEN PILE FOUNDATIONS

Geotechnical Information

The Lead Structural Engineer should request the design information from the geotechnical consultant as follows:

- Recommended pile types, axial, and lateral load capacities (including factors of safety used), and lengths.
- Estimated maximum settlement of pile foundations, with special consideration for pile groups under large or unusual structures.
- A method for determining the group efficiency for pile groups and the minimum spacing beyond which piles can be assumed to be behaving individually.
- Minimum recommended pile spacing.
- Passive resistance from soil on pile cap.
- Elastic spring constants (both horizontal and vertical) to be used for pile cap design with finite element programs.
- Lateral load versus lateral deflection (P-Y Curves) for the recommended pile types using pinned and/or fixed head conditions.
- Magnitude and location of the maximum moment in the pile to be used for structural design.
- Driving recommendations including the hammer type and size (required driving energy), refusal criteria, driving stresses to be considered for pile structural design, and special requirements such as pre-drilling or jetting.

PILE CAPACITIES

The load capacities of piles are derived from soil information provided by a geotechnical consultant and from the structural capacity of the pile itself. The capacity recommendations of the geotechnical consultant are usually made without specific consideration of the structural capacity of the pile. It is usually left to the Structural Engineer to ensure that the pile can sustain the loads to which it will be subjected. Design pile capacities given by the geotechnical consultant are estimates of the actual capacities which vary from pile to pile on any given site. Pile capacities are usually verified with test piles, especially where required by building codes. Building codes require static load pile tests; however, in some cases, the geotechnical consultant may recommend wave analysis of some kind.

A discussion of the soil parameters affecting pile capacities is provided below. These parameters include axial capacity (compression and tension), lateral capacity, group efficiency, settlement, and negative skin friction. There are various methods (and



DRIVEN PILE FOUNDATIONS

computer programs) for computing the capacities of piles given the soil parameters for a particular site. These methods may be found in the texts listed in the References section of this document. Lead Engineers may find these helpful in verifying information from the geotechnical consultant or to derive preliminary capacities to facilitate the start of a project.

Axial Capacity

Axial loads are transferred from the pile to the soil by friction, end bearing, or a combination of the two. The geotechnical consultant will state whether piles are friction or end bearing piles or to what extent friction and bearing are combined to get the total axial capacity. Uplift loads are transferred from the pile to the soil by friction only; however, the weight of the pile is added to the frictional resistance for the total tensile capacity of the pile.

Lateral Capacity

Lateral loads are transferred from the pile to the soil by passive pressure of the pile bearing against the soil. The lateral capacity of a pile is usually constrained by an allowable amount of lateral deflection and the strength of the soil on which the pile bears (laterally). In order for the geotechnical consultant to provide lateral pile capacity, the Lead Engineer should inform the consultant of the magnitude and character of the lateral loads to be sustained (wind, earthquake, hydrostatic, or gravity) for the project. In addition to the lateral capacity of the recommended pile, the geotechnical consultant will define the relationship between the applied lateral load and the lateral deflection to be experienced by the recommended piles for both free and fixed head conditions. This information can be used to establish the horizontal and rotational elastic spring constants needed to design the pile cap using finite element software.

Pile Group Efficiency

When piles are driven close together in groups, it is common that the load capacity of the group is less than the sum of the capacities of the individual piles within the group. This group efficiency is usually expressed as a percentage of the sum of the individual capacities of the piles within the group at a minimum pile spacing. The geotechnical consultant will define the efficiency of pile groups and/or a method of computation for group efficiency. In addition, the minimum pile spacing at which group effects may be ignored should be provided.

Usually, group efficiency is applicable only for sustained loading conditions such as dead load, operating load, and thermal load. This should be verified with the geotechnical consultant



DRIVEN PILE FOUNDATIONS

Settlement of Pile Groups

Settlement of pile groups will be investigated by the geotechnical consultant. This investigation should be made with a knowledge of the magnitude and character of the sustained loading to which the pile groups will be subjected. The settlement information will be used to evaluate the impact of differential settlement between adjacent structures as well as to define the vertical elastic spring constants used to design the pile cap if finite element software is used.

Negative Skin Friction

Negative skin friction (downdrag) occurs when fill is placed on a compressible soil deposit and a pile is driven into the compressible material before consolidation is complete. The principal result is an increase in the axial load on the pile, thereby decreasing its axial capacity. The geotechnical consultant will determine whether this situation exists and will provide design recommendations.

PILE DESIGN

Vertical Loading

The following equations can be used for the calculation of the pile loads assuming a rigid pile cap.

For rectangular pile patterns:

$$P = \frac{V}{n} \pm \frac{M_y X}{\sum X^2} \pm \frac{M_x Y}{\sum Y^2}$$

For circular, triangular, or other pile patterns:

$$P = \frac{V}{n} \pm \frac{Mc}{I}$$

where

P = Vertical load on any given pile

V = Total vertical load acting at the centroid of the pile group

n = Total number of piles

M = Moment about any axis through the centroid

M_x = Moment about the X-axis through the centroid



DRIVEN PILE FOUNDATIONS

M_y = Moment about the Y -axis through the centroid

X = Distance from the Y -axis to the pile in question

Y = Distance from the X -axis to the pile in question

ΣX^2 = Sum of the squares of the distances to each pile from the Y -axis

ΣY^2 = Sum of the squares of the distances to each pile from the X -axis

c = Distance from centroid of group to the pile in question

I = Moment of inertia of the pile group

$$= \frac{1}{8} \sum_{i=1}^j n_i d^2 \quad \text{for circular pile patterns}$$

where

d_i = Diameter of pile circle

n_i = Total number of piles in a circle whose diameter is d_i

j = Number of pile circles

Also, finite element software (plate elements on elastic supports) is available for the calculation of pile reactions and the design of pile caps. Application of finite elements for pile cap analysis and design should be performed under the direction of the Lead Structural Engineer with recommendations from the geotechnical consultant.

Lateral Loading

The allowable lateral load capacity of the pile group should be determined and compared to the actual lateral load. If the allowable load capacity is the greater, the foundation should be designed as a pile group with vertical loads only. If the actual lateral load is the greater, additional piles or battered piles may be used to carry the excess lateral loads. If recommended by the geotechnical consultant, passive resistance of the soil to the footing may be considered.

Battered Piles

Battered piles carry horizontal loads combined with vertical loads axially, reducing the need for large lateral pile capacity. In most cases, the analysis of battered pile groups is statically indeterminate. A method for determining the pile reactions for a group of battered piles is presented in Attachment 06, Sample Design 3. Other methods including finite elements (combination of plate elements and beam elements) are available and may be used at the discretion of the Lead Structural Engineer. Note that most manual methods



DRIVEN PILE FOUNDATIONS

assume that the piles are axially loaded, which precludes bending moments being developed. Some accounting for bending should be made even for battered piles.

Battered piles should have a maximum batter of 3 to 12 (horizontal and vertical respectively) and a minimum practical batter of 1 to 12. Piles with a batter greater than 3 to 12 will require special driving equipment and thereby increase driving costs. When using battered piles, avoid interference with other piles.

Structural Capacity

Once the soil capacities of a particular type of pile are established, the pile must be designed structurally to sustain these capacities. The structural capacity of a pile is based on the allowable stress for the material used (concrete, steel, or timber). The points of maximum stress are determined from information provided by the geotechnical consultant. In addition, the pile must be able to withstand handling and driving stresses.

There are special requirements for piles in UBC (Uniform Building Code) Seismic Zones 3 and 4. The UBC requires that piles will be designed for flexure whenever the tops of such members will be displaced by earthquake motions. The Design Engineer will follow the requirements of UBC in this case.

Allowable stresses in piles are usually specified in the building code that is enforced by the governing municipality where the piles are to be driven. Where building codes do not specify allowable stresses, concrete piles should be designed by the design procedures of the latest ACI (American Concrete Institute) 318 Building Code. Steel piles will meet the requirements of the latest AISC Manual. Timber piles should be designed to meet the requirements of the latest AITC (American Institute of Timber Construction) Manual.

PILE CAP DESIGN

Pile caps should be designed by the concrete design procedures of the latest ACI 318 Building Code. Rigid pile cap designs are presented in the sample designs contained in this document. Finite element solutions are available and can be used with the recommendation or approval of the Lead Structural Engineer. In addition, finite element models are extremely helpful for analyzing pile caps and load distribution to piles when field construction problems occur.

Minimum recommendations for pile cap design are as follows:

- Minimum edge distance should be 1'- 6" from the center of the pile to the face of concrete.
- Minimum embedment of piles into the cap should be 4 inches.
- Minimum thickness for the pile cap, based on ACI 15.7, should provide for 12 inches of concrete above the bottom reinforcing bars, with incremental increases of 1 inch.



DRIVEN PILE FOUNDATIONS

- Minimum bottom reinforcement will be #5 bars at 12 inches center-to-center both ways. The reinforcement will have 3 inches clearance above the tops of the piles.
- Minimum top reinforcement of #5 bars at 12 inches center-to-center both ways should be provided in foundations containing piles with uplift connectors.

Permanent eccentricities on pile groups can be a design consideration. The Design Engineer should consult the Lead Structural Engineer for considerations and recommendations for allowable permanent eccentricities.

Three pile foundations that utilize a triangular pile cap design for T-supports or other single column supports (refer to Attachment 08, Sample Design 5: Three Pile Foundation For T-Support) may require excessive construction man-hours due to the complex form work and rebar placement. A square pile cap should be considered and used if determined to be more economical than a triangular pile cap.

CONNECTIONS

The connections discussed below should be designed in accordance with the latest ACI 318 Building Code for concrete, the AISC Manual for steel, and the AITC Manual for timber.

Cap Plates Or Shear Tabs For H-Piles

Cap plates or shear tabs are required at the head of steel H-piles and hollow steel pipe piles to provide sufficient bearing area at the concrete to steel interface in the pile cap. The bearing area provided should be sufficient to transfer the pile axial capacity in both compression and tension. Cap plates are used when sufficient pile projection is available to allow field welding. Shear tabs are used where field welding is not feasible.

Pile Embedment

The minimum pile embedment given above is usually sufficient to provide a pinned head condition for a pile.

The embedment length required for fixed head piles is based on the lateral capacity of the pile. The shear and the induced bending moment at the head of the pile must be transferred from the pile cap to the pile in bearing. An assumed bearing pressure distribution is shown in Attachment 01, Figure 1. The allowable bearing stresses in the concrete pile cap, as well as the pile, should not be exceeded. Usually, the bearing capacity of the concrete for the pile cap will control. The pile must be designed to carry the shear and bending stresses along the full length of the pile embedment.



DRIVEN PILE FOUNDATIONS

Tension Connections

Tension connections for piles are designed based on the development length of bars in concrete according to ACI 318 Building Code or by shear cone theory similar to anchor bolt and headed stud design. Some typical tension connection details are presented in Attachments 1, 2, and 3.

Pile Splices

Concrete piles are usually cast to the required length; however, splices can be made with steel inserts grouted into holes made during casting. Long steel and timber piles require splices since these materials are usually in stock lengths. In general, steel pile splices involve welded flange and web plates.

Splices for timber piles usually require the connected ends to be tapered with bearing surfaces finished. The tapered ends are held together by a steel ring. Some typical splice details are presented in the References section of this document.

Special Requirements for Seismic Zones

Connections designed for seismic loads must meet the ductile moment resisting frame requirements. Mechanical fasteners for concrete piles should not be used to resist seismic forces.

REFERENCES

ACI (American Concrete Institute)

AISC (American Institute for Steel Construction)

AITC (American Institute of Timber Construction)

UBC (Uniform Building Code)

Bowles, Joseph E. Foundation Analysis and Design, 3rd ed. McGraw-Hill, Inc. New York. 1982.

Davisson, M. T., and H. L. Gill. "Lateral Loaded Piles In A Layered Soil System." Journal of the Soil Mechanics and Foundations Division. ASCE (American Society of Civil Engineers). May 1963.

Gaylord, Edwin H., Jr., and Charles N. Gaylord. Structural Engineering Handbook, 2nd ed. McGraw-Hill, Inc. New York. 1979.

Peck, Ralph B., Walter E. Hanson, and Thomas H. Thornburn. Foundation Engineering, 2nd ed. John Wiley & Sons, Inc. New York. 1974.

DRIVEN PILE FOUNDATIONS

Teng, Wayne C. Foundation Design. Prentice Hall, Inc. Englewood Cliffs, NJ. 1962.

Winterkorn, Hans F., and Hsai-Yang Fang. Foundation Engineering Handbook. Van Nostrand Reinhold Company. New York. 1975.

Structural Engineering

Guideline 000.215.1207: Anchor Bolt Design Criteria

Structural Engineering

Guideline 000.215.1220: Vertical Vessel Foundations

Structural Engineering

Guideline 000.215.1221: Heat Exchanger Foundations

Structural Engineering**Guideline 670.215.1222:** Horizontal Vessel Foundations

Structural Engineering

Guideline 000.215.1223: Sphere Foundations

Structural Engineering

Guideline 000.215.1224: Heater Foundations

Structural Engineering

Guideline 000.215.1225: Combined Foundations

Structural Engineering

Guideline 000.215.1231: Drilled Pile Foundations

Structural Engineering

Guideline 000.215.1250: Pipe Supports

ATTACHMENTS**Attachment 01:**

Figure 1. Assumed Pressure Distribution For Fixed Headed Pile Embedment

Figure 2. Tension Connector For Prestressed Concrete Pile

Attachment 02

Figure 1. Tension Connector For Concrete Filled Steel Pile

Figure 2. Tension Connector For Steel Pile

Attachment 03:

Timber Pile Uplift Connector

Attachment 04:

Sample Design 1: Vertical Vessel Foundation On Piles

Attachment 05

Sample Design 2: Heat Exchanger Foundation On Piles With A Grade Beam



DRIVEN PILE FOUNDATIONS

Attachment 06

Sample Design 3: Piperack Braced Bay Foundation On Battered Piles

Attachment 07:

Sample Design 4: Combined Foundation On Piles

Attachment 08:

Sample Design 5: Three Pile Foundation For T-Support

Attachment 09:

Sample Design 6: Sphere Foundation On Piles

Attachment 10:

Sample Design 7: Cylindrical Heater Foundation On Piles

Attachment 11:

Sample Design 8: Structural Design Of A Prestressed Concrete Pile

DRIVEN PILE FOUNDATIONS

Figure 1. Assumed Pressure Distribution for Fixed Headed Pile Embedment

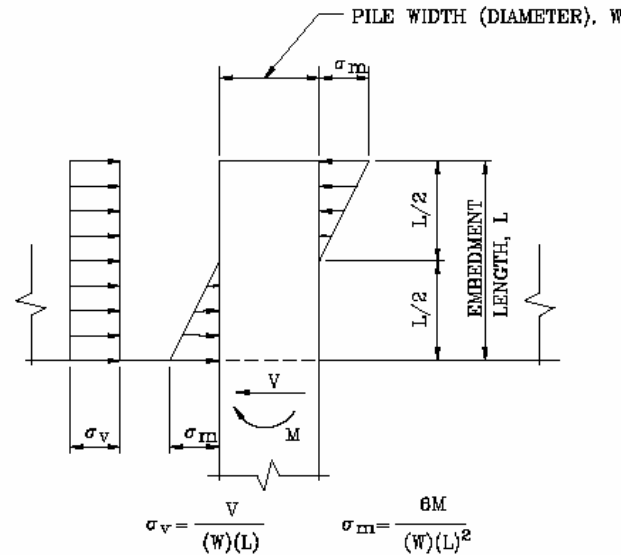
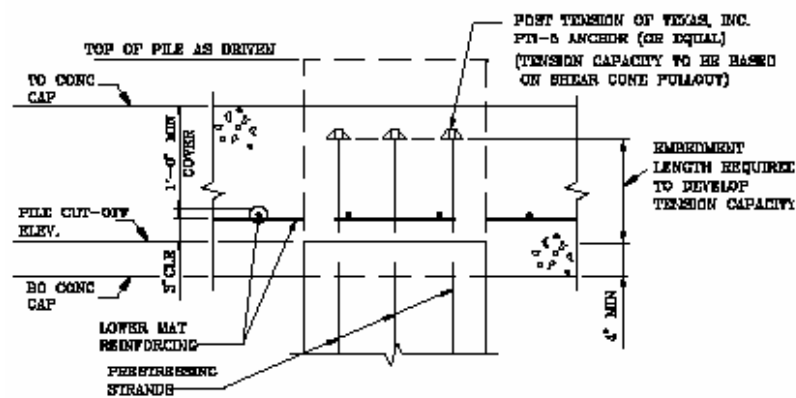


Figure 2. Tension Connector for Prestressed Concrete Pile



- Notes:
1. PTI-5 anchors must be mechanically fixed to prestressing strand prior to pouring pile cap.
 2. PTI-5 anchors have been tested for cyclical dynamic loading.

DRIVEN PILE FOUNDATIONS

Figure 1. Tension Connector for Concrete Filled Steel Pipe

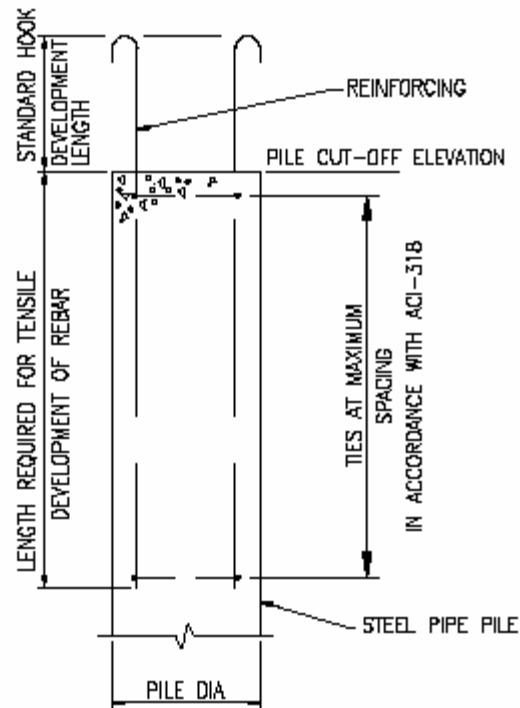
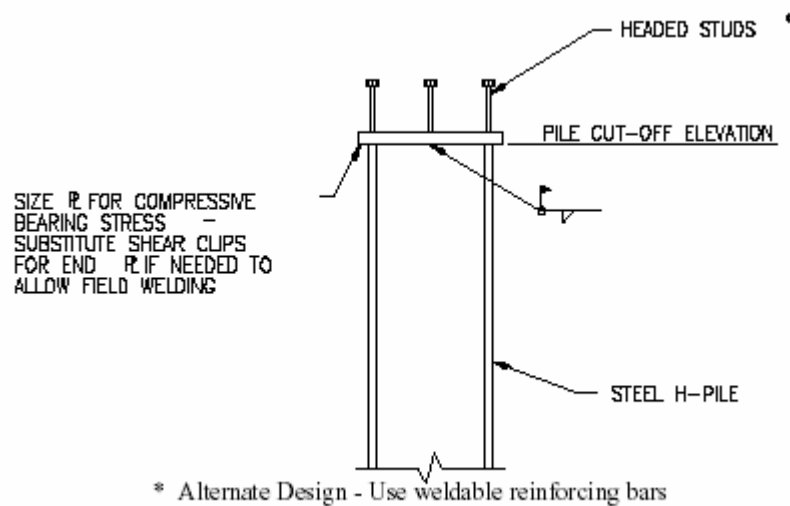
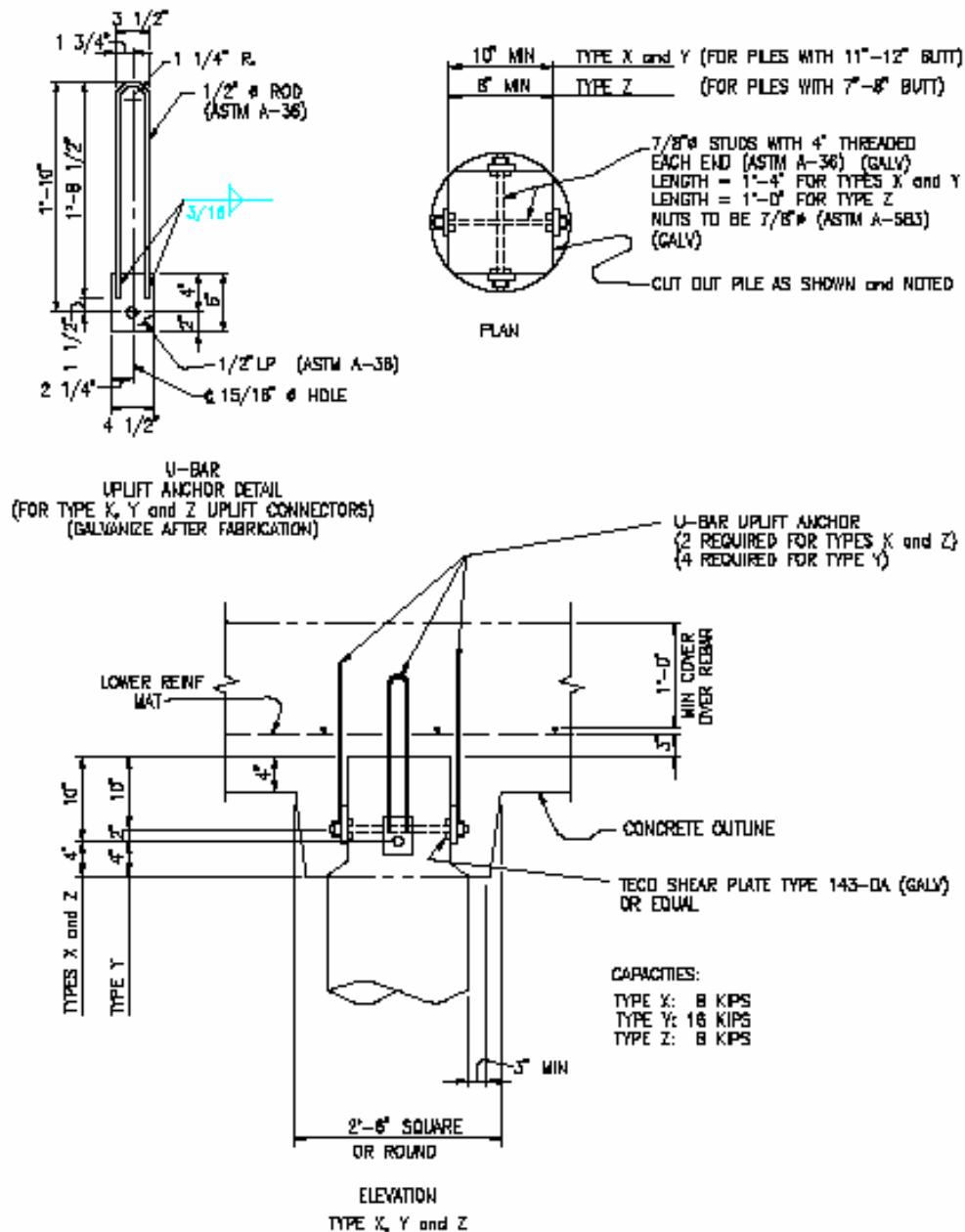


Figure 2. Tension Connector for Steel Pipe



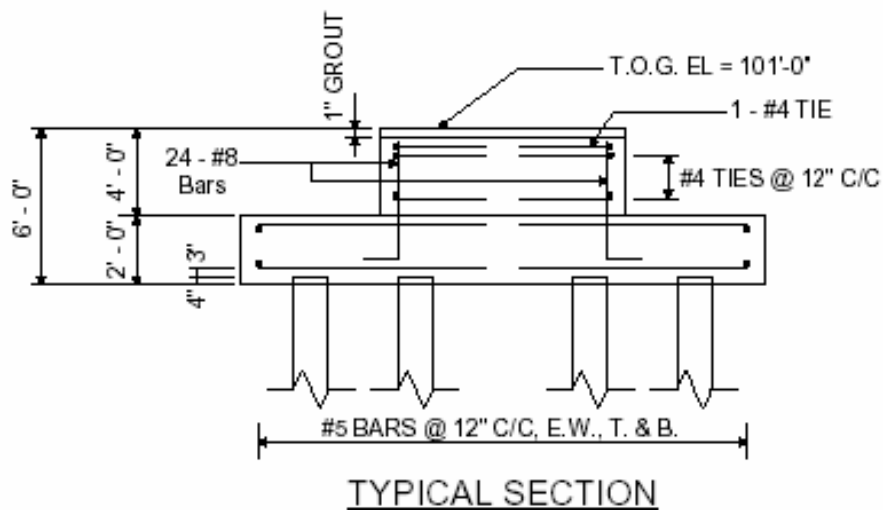
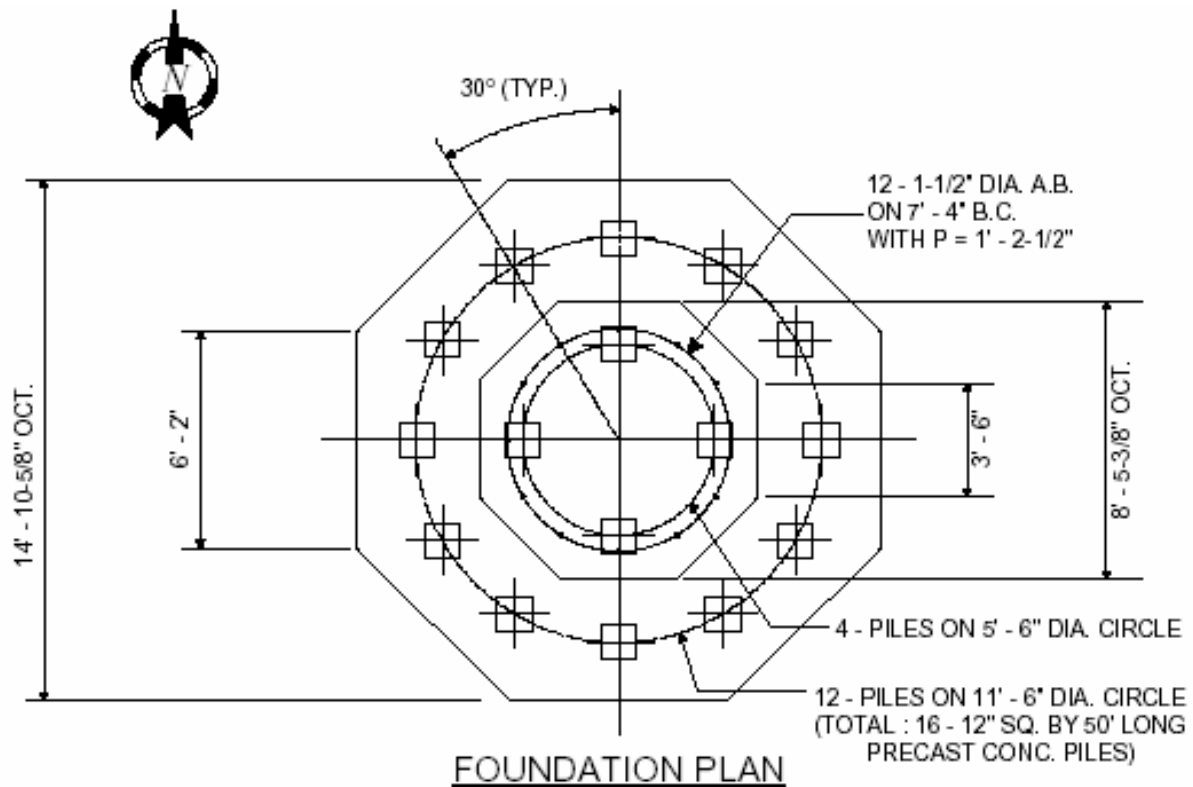


TIMBER PILE UPLIFT CONNECTOR





SAMPLE DESIGN 1: VERTICAL VESSEL FOUNDATION ON PILES





SAMPLE DESIGN 1: VERTICAL VESSEL FOUNDATION ON PILES

DESIGN DATA

References

ACI-318-89 for concrete design
Vessel data sheet and load calculations
Vendor drawings

Materials

Concrete : $f'_c = 3,000$ psi, $\gamma_{conc} = 150$ pcf

Rebar : ASTM A-615 Grade 60

Piles : Prestressed Concrete : 12" square x 50' long (Free Head)
Minimum Spacing = 3'-0" c/c
Allowable Compression = 60 k
Allowable Tension = 30 k
Allowable Shear = 2 k
Allowable Group Capacity = $Q_d = [PL_c + 2.5q_u(1 + 0.3w/b)A] / FS$
(See calculations for definition of variables)

Anchor Bolts : 12 - 1 1/2" diameter anchor bolts on 7'-4" dia. bolt circle

Soil : $\gamma = 100$ pcf
Average Cohesion, C, in top 55' of soil = 700 psf
Average Unconfined Compressive Strength, q_u , of soil 20' below pile tips = 2,000 psf

Design Loads

Empty Weight = 121 k
Operating Weight = 200 k
Test Weight = 504 k
Wind Loads at base of vessel base plate: Shear = 14 k
Moment = 884 ft-k

PEDESTAL DESIGN

Note : For brevity, pedestal design is not shown here. See Guideline 000-215-1220, Vertical Vessel Foundations, for details of pedestal design.

Use 8' - 5 3/8" Octagon ($A = 59.15$ ft²)

Try 4' - 0" pedestal height (2' - 0" Pile cap thickness; Check cap thickness for development of dowels per ACI-318)



SAMPLE DESIGN 1: VERTICAL VESSEL FOUNDATION ON PILES

PILE CAP DESIGN

Preliminary Size

(This section provided for information only. Do not provide in actual calculations. Do state trial size.)

Use trial and error method to determine number of piles required, with the following considerations :

$$\text{Number of piles required for shear} = \text{WIND SHEAR} / \text{ALLOWABLE SHEAR} = 14 / 2 = 7$$

$$\begin{aligned} \text{Number of piles required for test} &= \frac{\text{TEST WT.} + (\text{AREA PED.})(\text{HEIGHT})(\gamma_{\text{conc}})}{(\text{ALLOW. COMP.})(\text{GROUP EFFICIENCY})} \\ &= [504 + 59.15 \times 6 \times 0.15] / [60 \times 0.80 \text{ \{assumed\}}] = 12 \end{aligned}$$

Number of piles required for operation + wind

$$\begin{aligned} P_{\text{max/min}} = 60 \text{ k} &= V / N \pm M_c / I ; \text{ where } V = 200 + \text{FND. WT.} \\ &= 200 + (\text{AREA PILE.CAP.})(\text{DEPTH})(\gamma_{\text{avg}}) \\ &\text{where } \gamma_{\text{avg}} = (\gamma_{\text{conc}} + \gamma_{\text{soil}}) / 2 \\ M &= 884 \text{ ft-k} + 14 \times \text{DEPTH} = 884 + 14 \times 6' = 968 \text{ ft-k} \end{aligned}$$

$$\begin{aligned} \text{Try 12 Piles : } 4 @ 3 \sqrt{2} &= 4' - 3" \text{ dia.} \\ 8 @ 4.25 + 2 \times 3 &= 10' - 3" \text{ dia.} \end{aligned}$$

$$\text{Octagon} = 10.25 + 3 = 13' - 3"$$

$$V = 200 + 0.83 \times (13.25)^2 \times 5 \times 0.125 = 291 \text{ k}$$

$$I = \frac{1}{8} [8(10.25)^2 + 4(4.25)^2] = 114 \text{ ft}^2$$

$$P_{\text{max/min}} = 291 / 12 \pm 968(5.13) / 114 = 24 \pm 44 = 68 \text{ k, } -20 \text{ k} > 60 \text{ k (not good)}$$

Since "Moment " reaction controls, increase Section Modulus.

$$1 / S = c / I = (60 - 24) / 968 = 0.037 \rightarrow S = 27'$$

$$\begin{aligned} \text{Try 14 Piles : } 4 @ 3 \sqrt{2} &= 4' - 3" \text{ dia.} \\ 10 @ 10' - 3" &\text{ dia.} \end{aligned}$$

$$S = 1/8 [10(10.25)^2 + 4(4.25)^2] / 5.13 = 27.4' > 27' \text{ (too close)}$$



SAMPLE DESIGN 1: VERTICAL VESSEL FOUNDATION ON PILES

Try 16 Piles : 12 @ 3 x 12 / π = 11.5' dia.
 4 @ 11.5 - 6 = 5.5' dia.

$$V = 200 + 0.83(14.5)^2 \times 5 \times 0.125 = 309 \text{ k}$$

$$I = \frac{1}{8} [12(11.5)^2 + 4(5.5)^2] = 214 \text{ ft}^2$$

$$P_{\max/\min} = 309 / 16 \pm 968(5.75) / 214 = 20 \pm 26 = 46 \text{ k}, -6 \text{ k} < 60 \text{ k}, -30 \text{ k (okay)}$$

Try 11.5 + 3 = 14.5; Say 14' - 10 $\frac{5}{8}$ " Octagon w/ 12 - Piles @ 11' - 6" dia. and 4 - Piles @ 5' - 6" dia.
Thickness = 2' - 0"

Foundation Weight

$$\begin{aligned} \text{Pedestal Weight} &= 59.15 \times 4 \times 0.15 = 35.5 \text{ k} \\ \text{Pile Cap Weight} &= 183.62 \times 2 \times 0.15 = 55.1 \text{ k} \\ \text{Soil Weight} &= (183.62 - 53.64)3 \times 0.1 = 39.0 \text{ k} \\ \text{Total} &= \underline{129.6 \text{ k}} \end{aligned}$$

I of Pile Group

$$I = \frac{1}{8} [12(11.5)^2 + 4(5.5)^2] = 213.5 \text{ ft}^2$$

Check Pile Spacing

$$\begin{aligned} 11' - 6" \text{ dia. Circle : } & S = d_i \sin (180^\circ / n_i) = 11.5 \sin (180^\circ / 12) = 3.0' = 3' \text{ (okay)} \\ 5' - 6" \text{ dia. Circle : } & S = 5.5 \sin (180^\circ / 4) = 3.9' > 3' \text{ (okay)} \end{aligned}$$

Group Capacity

To be checked against Test Load only

$$Q_d = [\pi(11.5 + 1.0)(50)(0.70) + 2.5(2.0)(1 + 0.3 \times 12.5 / 12.5)(\pi/4 \times (12.5)^2)] / 3 = 724 \text{ k}$$

where O.D. of Pile Group = 11.5 + 1.0 = 12.5'
 P = Perimeter of Pile Group
 L = Length of Piles
 C = Average Cohesion
 q_u = Average Unconfined Compressive Strength
 w = Width of Pile Group (dia. in this case)
 b = Length of Pile Group (dia. in this case)
 A = Pile Group Area
 FS = Factor of Safety

$$\text{Reduced Pile Capacity} = 724 / 16 = 45.3 \text{ k}$$



SAMPLE DESIGN 1: VERTICAL VESSEL FOUNDATION ON PILES

Group Efficiency = $45.3 / 60 = 75.5\%$

Check Pile Loads

Case I : Empty + Wind

$V = \text{EMPTY WT.} + \text{FND. WT.} = 121 + 129.6 = 250.6 \text{ k}$

$P_{\max/\min} = V / N \pm Mc / I = 250.6 / 16 \pm 968(5.75) / 213.5 = 15.7 \pm 26.1 = 41.8 \text{ k}, -10.4 \text{ k} < 60 \text{ k}, -30 \text{ k}(\text{okay})$

Shear / Pile = $14 / 16 = 0.9 \text{ k} / \text{Pile} < 2 \text{ k}(\text{okay})$

Case II : Operating + Wind

$V = 200 + 129.6 = 329.6 \text{ k}$

$P_{\max/\min} = 329.6 / 16 \pm 26.1 = 20.6 \pm 26.1 = 46.7 \text{ k}, -5.5 \text{ k} < 60 \text{ k}, 30 \text{ k}(\text{okay})$

Shear / Pile - Same as Case I

Case III : Test + 30 mph Wind (Design Wind = 100 mph)

$V = 504 \text{ k} + 129.6 = 633.6 \text{ k}$

$P_{\max/\min} = 633.6 / 16 \pm 26.1 \times (30)^2 / (100)^2 = 39.6 \pm 2.3 = 41.9 \text{ k}, 37.3 \text{ k} < 45.3 \text{ k}(\text{okay})$

Shear / Pile - Okay by Inspection

Check 5' - 6" dia. circle for uplift in Case I

$P_{\min} = 15.7 - 968(2.75) / 213.5 = 3.2 \text{ k}(\text{No Uplift})$

Tension connectors not required on 5' - 6" dia. circle

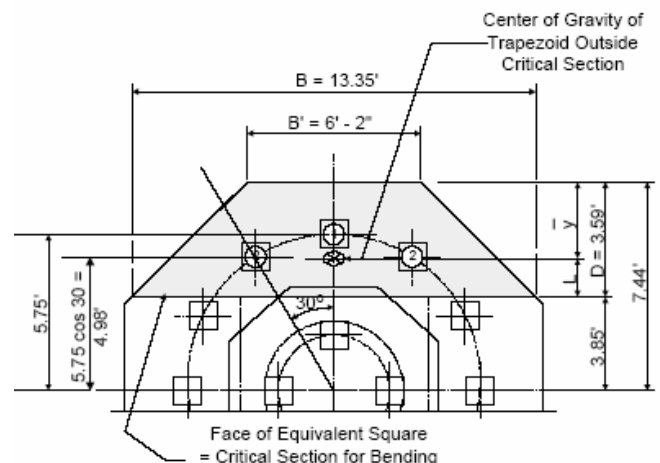
Reinforcement Design

Equivalent Sq. Dimension for Pedestal = $\sqrt{59.15} = 7.69'$

Area Trapezoid = $(B + B') \times (D / 2)$
 $= (13.35 + 6.17) \times 3.59 / 2$
 $= 35.1 \text{ ft}^2$

$y = (D / 3) \times [(2B + B') / (B + B')]$
 $= (3.59 / 3) \times (2 \times 13.35 + 6.17) / (13.35 + 6.17)$
 $= 2.02'$

$L = 3.59 - 2.02 = 1.57'$





SAMPLE DESIGN 1: VERTICAL VESSEL FOUNDATION ON PILES

Soil and Concrete Weight on
Trapezoid = $35.1(0.1 \times 3 + 0.15 \times 2) = 21.1 \text{ k}$
Bottom Steel : Shear and Moment @ Face of Equivalent Square

Case I :

$$P_1 = 0.9 \times 15.5 + 1.3 \times 26.1 = 47.9 \text{ k}$$

$$P_2 = 0.9 \times 15.7 + 1.3(968 \times 4.98 / 213.5) = 43.3 \text{ k}$$

$$V_u = 47.9 + 2 \times 43.3 - 0.9 \times 21.1 = 115.5 \text{ k } (\div 13.35 = 8.7 \text{ k / ft})$$

$$M_u = 47.9 \times 1.90 + 2 \times 43.3 \times 1.13 - 0.9 \times 21.1 \times 1.57 = 159 \text{ ft-k } (\div 13.35 = 11.9 \text{ ft-k / ft})$$

Case II :

$$P_1 = 0.75(1.4 \times 20.4 + 1.7 \times 26.1) = 54.7 \text{ k}$$

$$P_2 = 0.75(1.4 \times 20.6 + 1.7 \times (968 \times 4.98 / 213.5)) = 50.2 \text{ k}$$

$$V_u = 54.7 + 2 \times 50.2 - 0.75 \times 1.4 \times 21.1 = 132.9 \text{ k } (\div 13.35 = 10.0 \text{ k / ft})$$

$$M_u = 54.7 \times 1.9 + 50.2 \times 1.13 - 0.75 \times 1.4 \times 21.1 \times 1.57 = 182.6 \text{ ft-k } (\div 13.35 = 13.7 \text{ ft-k / ft})$$

Case III : Without Wind

$$P_1 = P_2 = 1.4 \times 39.4 = 55.2 \text{ k}$$

$$V_u = 3 \times 55.2 - 1.4 \times 21.1 = 136.1 \text{ k } (\div 13.35 = 10.2 \text{ k / ft}) \leftarrow \text{CONTROLS}$$

$$M_u = 55.2(1.9 + 2 \times 1.13) - 1.4 \times 21.1 \times 1.57 = 183.3 \text{ ft-k } (\div 13.35 = 13.7 \text{ ft-k / ft}) \leftarrow \text{CONTROLS}$$

Steel Required

$$d = 24 - 4 - 3 - 1.5(1" \text{ bar dia.}) = 15.5"$$

$$F = bd^2 / 12,000 = 12(15.5)^2 / 12,000 = 0.240$$

$$K_u = M_u / F = 13.7 / 0.24 = 57.1 \rightarrow a_u = 4.45$$

$$A_s = M_u / (a_u \times d) = 13.7 / (4.45 \times 15.5) = 0.20 \text{ in.}^2 / \text{ft } (\times \frac{4}{3} = 0.27 \text{ in.}^2 / \text{ft}) \leftarrow \text{CONTROLS}$$

$$A_s @ \rho_{\min} = 0.0033(15.5)(12) = 0.61 \text{ in.}^2 / \text{ft}$$

SAMPLE DESIGN 1: VERTICAL VESSEL FOUNDATION ON PILES

USE #5 bars @ 12" c/c E.W. (A_s provided = 0.31 in.² / ft)

Check Beam Shear

$$V_c = \phi 2 \sqrt{f'_c} \times b_w \times d = 0.85 \times 2 \times \sqrt{3,000} \times 12 \times 15.5 = 17.3 \text{ k} > 10.2 \text{ k (okay)}$$

Top Steel

Case I is critical (By Inspection)

$$P_1 = 0.9 \times 15.5 - 1.3 \times 26.1 = -20.0 \text{ k}$$

$$P_2 = 0.9 \times 15.5 - 1.3 \times (968 \times 4.98 / 213.5) = -15.4 \text{ k}$$

$$V_u = 20.0 + 2 \times 15.4 + 0.9 \times 21.1 = 69.8 \text{ k } (\div 13.35 = 5.2 \text{ k / ft})$$

$$M_u = 20.0 \times 1.9 + 2 \times 15.4 \times 1.13 + 0.9 \times 21.1 \times 1.57 = 102.6 \text{ ft-k } (\div 13.35 = 7.7 \text{ ft-k / ft})$$

$$7.7 \text{ ft-k / ft} < 13.7 \text{ ft-k / ft} \text{ \& } d_{bot} < d_{top} \text{ (By Inspection)}$$

Use #5 bars @ 12" c/c E.W. Top (A_s provided = 0.31 in.² / ft)

Check Punching Shear

Case III controls

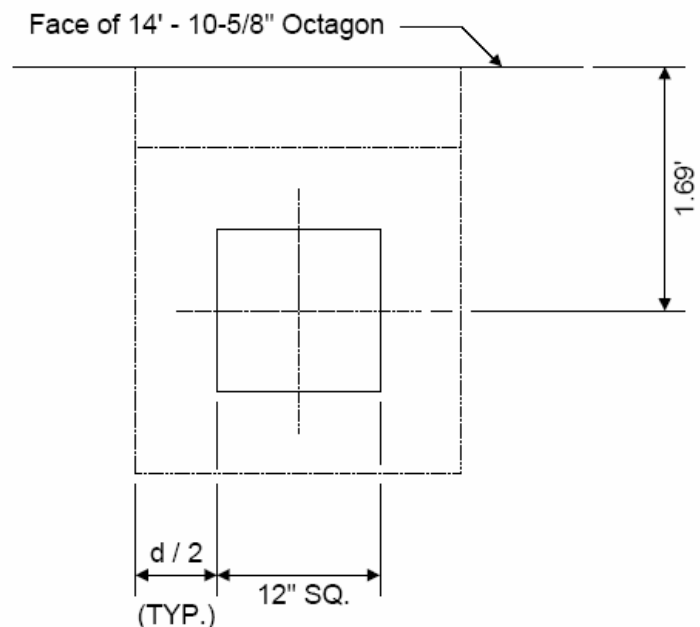
$$d = 15.5''$$

$$b_o = 4(b + d) = 4(12 + 15.5) = 110''$$

OR

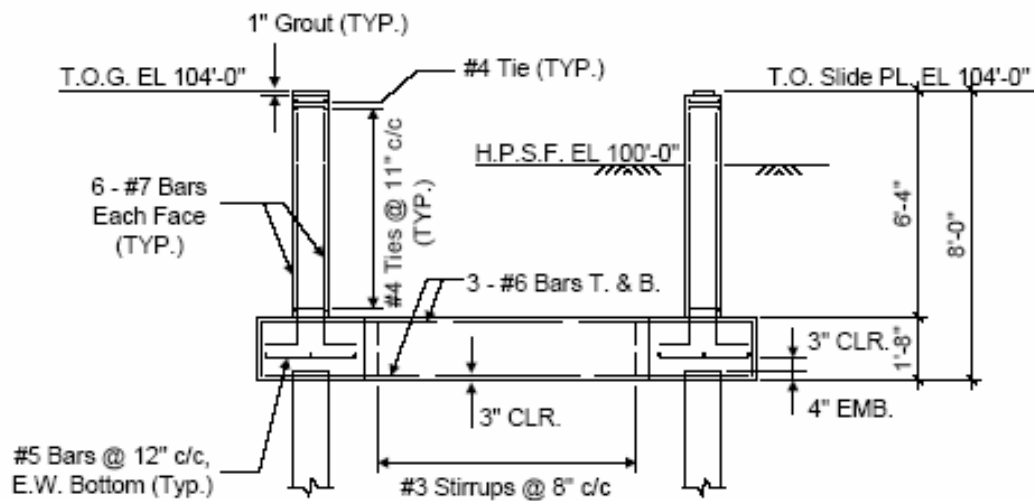
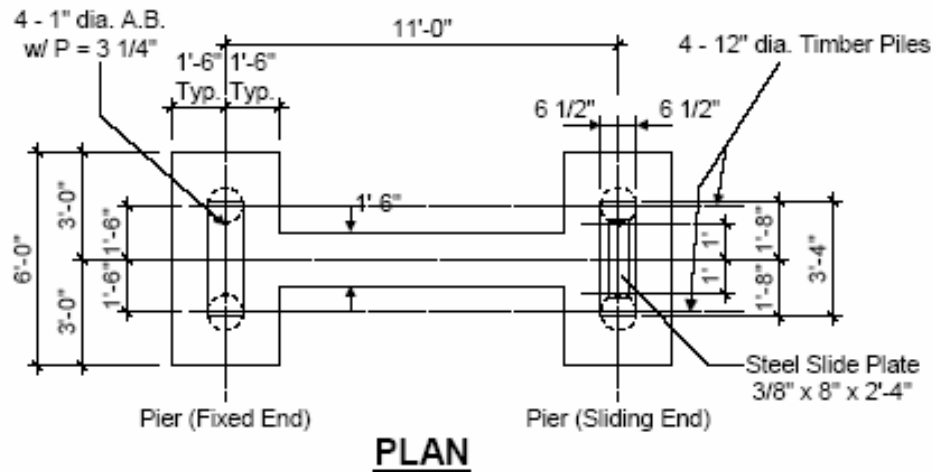
$$\begin{aligned} b_o &= 2(b + d + 1.69 \times 12) \\ &= 2(12 + 15.5 + 20.3) = 95.6'' \leftarrow \underline{\text{USE}} \end{aligned}$$

$$\begin{aligned} \phi V_n &= \phi 4 \sqrt{f'_c} \times b_o \times d \\ &= 0.85 \times 4 \times \sqrt{3,000} \times 95.6 \times 15.5 \\ &= 276 \text{ k} > 55.2 \text{ k (okay)} \end{aligned}$$





SAMPLE DESIGN 2: HEAT EXCHANGER FOUNDATION ON PILES WITH A GRADE BEAM





SAMPLE DESIGN 2: HEAT EXCHANGER FOUNDATION ON PILES WITH A GRADE BEAM

DESIGN DATA

References

ACI-318-89 for concrete design

Vendor drawings

Materials

Concrete : $f'_c = 3,000$ psi, $\gamma_{conc} = 150$ pcf

Rebar : $f_y = 60,000$ psi

Piles :	12" dia. Timber Piles (Free Head)	
	Minimum Spacing =	3' - 0"
	Allowable Compression =	30 T
	Allowable Tension =	15 T
	Allowable Shear =	1 T
	Minimum Edge Distance =	1' - 6"
	Minimum Embedment =	4"

} 1/3 increase in capacities for
load combinations that include
Seismic Loads or Wind Loads

Anchor Bolts : 4' - 1" dia. Anchor Bolts, A36 (2 Each Pier)

Soil : $\gamma_{soil} = 100$ pcf
Passive Resistance = 200 psf / ft of depth

Design Loads

Note : For brevity, design loads and load combinations are not shown here. See practice 670-215-1221, Heat Exchanger Foundations, for required design load information and derivations.

Governing load condition for each design item is presented with that item.

PIER DESIGN

Note : For brevity, pier design including : pier dimensions, anchor bolt design, slide plate design, and reinforcement are not shown here. See practice 670-215-1221, Heat Exchanger Foundations, for details of pier design.

Use 3' - 4" x 1' - 1" Pier, with Steel Slide Plate $\frac{3}{8}$ " x 8" x 2' - 6", and 6 - #7 Vertical Bars E.F., #4 Ties @ 11" c/c (Check development of Vertical Reinforcement in Pile Cap, per ACI-318.)

PILE CAP DESIGN

Try 6' - 0" x 3' - 0" x 1' - 8" Thick Pile Cap, with 2 Piles under each Pier



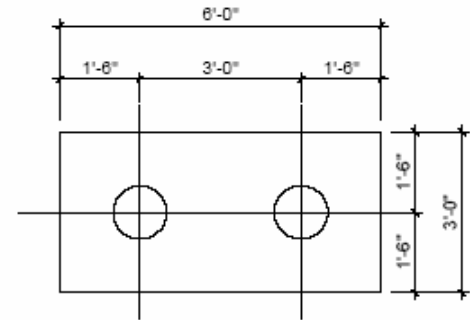
SAMPLE DESIGN 2: HEAT EXCHANGER FOUNDATION ON PILES WITH A GRADE BEAM

Pile Cap Weight

$$\begin{aligned}\text{Pier Weight} &= 3.33(1.08)(6.33)(0.15) = 3.4 \text{ k} \\ \text{Cap Weight} &= 6.0(3.0)(1.67)(0.15) = 4.5 \text{ k} \\ \text{Soil Weight} &= (18 - 3.61)(2.33)(0.10) = 3.4 \text{ k} \\ \text{Total} &= \underline{11.3 \text{ k}}\end{aligned}$$

(Assume Grade Beam Width = 1.5')

$$\begin{aligned}\text{Grade Beam Weight} &= 1.5(1.67)(11.0 - 3.0)(0.15) = 3.0 \text{ k} \\ \text{Soil Weight on Beam} &= 1.5(2.33)(8.0)(0.10) = 2.8 \text{ k} \\ \text{Total} &= \underline{5.8 \text{ k}}\end{aligned}$$



Check Pile Capacity

Operating Only

$$P = 63.4 + 11.3 + 5.8 / 2 = 77.6 \text{ (63.4 = Load at Top of Pier)}$$

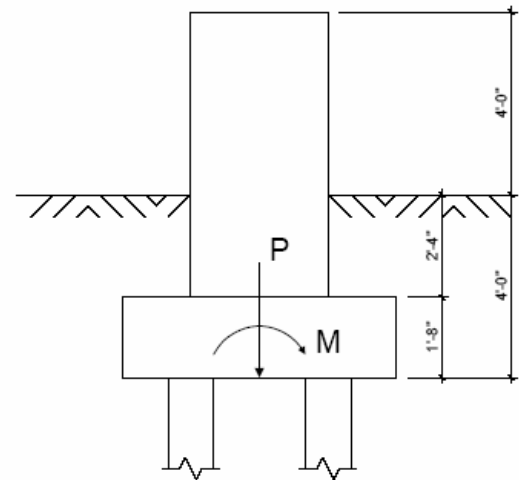
$$\text{Pile Load} = 77.6 / 2 = 38.8 \text{ k} < 60 \text{ k (okay)}$$

Operating + Seismic, Transverse Loading

$$P = 77.8 \text{ k}, M = 41.8 \text{ ft-k}, V = 7.6 \text{ k}$$

$$M @ \text{Base of Pile Cap} = 41.8 \text{ ft-k} + 7.6 \text{ k} (8.0 \text{ ft}) = 102.6 \text{ ft-k}$$

$$\begin{aligned}\text{Pile Load} &= P / N \pm M / S = 77.8 / 2 \pm 102.6 / 3 \\ &= 73.1 \text{ k}, 4.7 \text{ k} < 60 \times 1.33 = 80 \text{ k (No Uplift) (okay)}\end{aligned}$$



Check Lateral Resistance

$$V_{\max} = 7.6 \text{ k (Operating + Seismic, Transverse)}$$

$$\text{Passive Resistance} = 0.2[(4 + 2.33) / 2](3.0)(1.67) = 3.2 \text{ k (Base Only)}$$

$$\text{Passive Resistance} = 0.2(2.33 / 2)(1.08)(2.33) = 0.6 \text{ k}$$

$$\text{Pile Resistance} = 2(2) = 4.0 \text{ k}$$

$$\begin{aligned}\text{(Include 1/2 Grade Beam if necessary; however,} & \underline{0.0 \text{ k}} \\ \text{Gr. Bm. must then be designed for lateral loads)} & \end{aligned}$$

$$\text{Total} = \underline{7.8 \text{ k (x 1.33 = 10.4 k)}}$$

$$7.6 \text{ k} < 10.4 \text{ k (Pile Lateral Capacity - okay)}$$

Use Pile Capacity as described above with 2 Piles @ each Pier

Reinforcement Design

Since Piles are located directly under Piers, shear, moment, and puncture shear stresses in the Pile Caps are small. Provide minimum reinforcement.

Use #5 Bar @ 12" c/c, E.W., Bottom (No uplift; therefore, Top Steel not required)



SAMPLE DESIGN 2: HEAT EXCHANGER FOUNDATION ON PILES WITH A GRADE BEAM

GRADE BEAM DESIGN

Try 1' - 6" wide by 1' - 8" deep Gr. Bm. between Pile Caps

(Operating + Longitudinal Seismic controls)

$U = 1.05D + 1.4E$ (ACI 318-89, Section 9.2.3)

P_A : (Ultimate Load)

$P_{SH} + FDN. = 1.05(42.2 + 11.3) = 56.4 \text{ k}$

Seismic Couple = $1.4(6.4) = 9.0 \text{ k}$
Total = 65.4 k

P_B : (Ultimate Load)

$P_{CH} + FDN. = 1.05(63.4 + 11.3) = 78.6 \text{ k}$

Seismic Couple = $-1.4(6.4) = -9.0 \text{ k}$
Total = 69.4 k

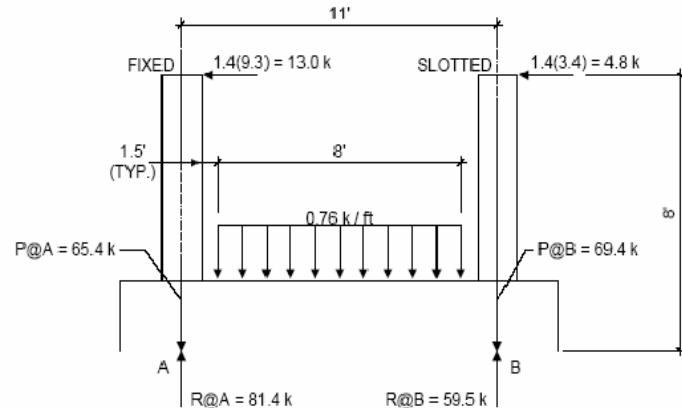
ΣM about Point A :

$11R_B + 8(13.0 + 4.8) - 11(69.4) - 0.76(8)(1.5 + 4) = 0$

$R_B = 654.4 / 11 = 59.5 \text{ k}$

$R_A = 65.4 + 69.4 + 0.76(8) - 59.5 = 81.4 \text{ k}$

$d = 20 - 3.5" = 16.5"$



Reinforcement Design

Flexural Reinforcement

$M_u = 95.3 \text{ ft-k}$, $b = 18"$, $d = 20 - 3 - 0.5 \times 2 = 16"$

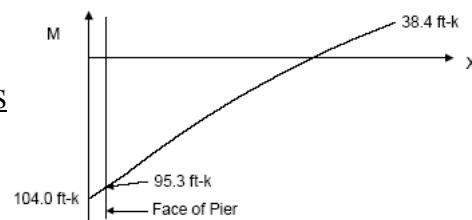
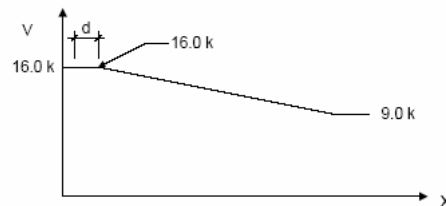
$F = 18(16)^2 / 12,000 = 0.384$

$K_u = 95.3 / 0.384 = 248.2 \rightarrow a_u = 4.24$

$A_s \text{ Required} = 95.3 / (4.24 \times 16) = 1.40 \text{ in.}^2 \leftarrow \text{CONTROLS}$

$A_s @ \rho_{\min} = 0.0033(18)(16) = 0.95 \text{ in.}^2 / \text{ft}$

Use 3 - #7 Bars Top & Bottom ($A_s \text{ Provided} = 1.80 \text{ in.}^2$)



Beam Shear

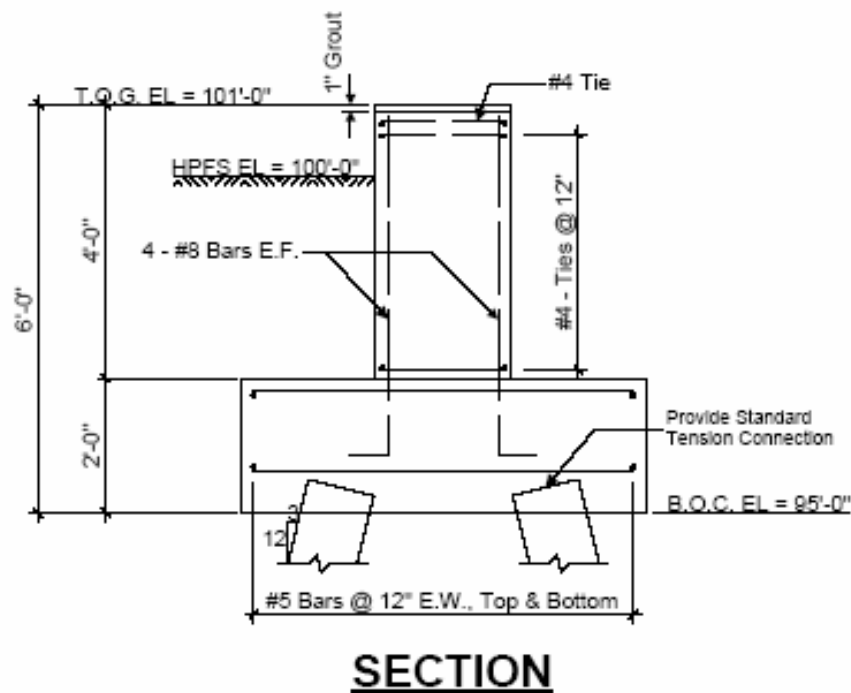
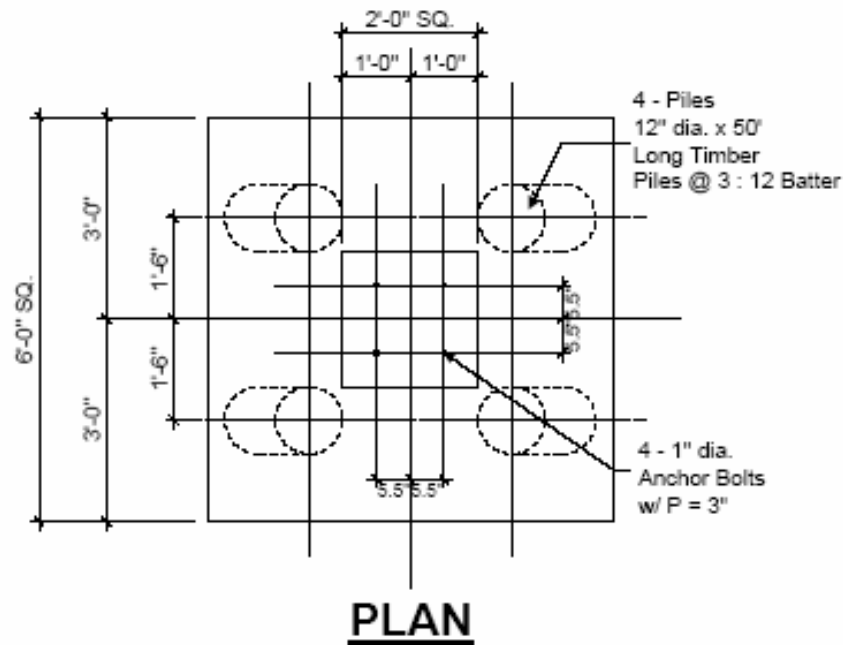
$V_u = 16.0 \text{ k} @ d$ from face of Pier

$\phi V_c = 2 \sqrt{3,000} (18)(16) / 1000 = 31.5 \text{ k} (\div 2 = 15.8 \text{ k} \approx 16 \text{ k})$

Minimum shear reinforcement required. $\rightarrow$ Use #3 Ties @ 8" c/c



SAMPLE DESIGN 3: PIPERACK BRACED BAY FOUNDATION ON BATTERED PILES





SAMPLE DESIGN 3: PIPERACK BRACED BAY FOUNDATION ON BATTERED PILES

DESIGN DATA

References

ACI-318-89 for concrete design

Piping drawings

Materials

Concrete : $f'_c = 3,000$ psi, $\gamma_{conc} = 150$ pcf

Rebar : ASTM A-615 Grade 60

Piles : 12" dia. Timber Piles
 Maximum Batter 3 : 12
 Minimum Spacing = 3' - 0" c/c
 Minimum Edge Distance = 1' - 6"
 Minimum Embedment = 4"
 Allowable Compression = 10 T, 20 T, 30 T
 Allowable Tension = 8T, 16 T, 24 T
 Allowable Shear = 1 T (for Wind only)

Thermal to be resisted by Battered Piles

Anchor Bolts : 4' - 1" dia. Anchor Bolts, A36 (2 Each Pier)

Soil : $\gamma_{soil} = 100$ pcf

Design Loads

Note : For brevity, design loads and load combinations are not shown here. See Guideline 000-215-1250, Pipe Supports, for required design load information and derivations.

Foundation is designed for operating gravity and Longitudinal Thermal Loads as follows :

$P_{oper} = 18$ k, $V_{thermal} = \pm 7$ k, $P_{thermal} = \pm 10.5$ k

PIER DESIGN

Note : For brevity, pier design including : pier dimensions, anchor bolt design, slide plate design, and reinforcement are not shown here. See Guideline 000-215-1250, Pipe Supports, for details of pier design.

Use 2' - 0" Square Pier, with 4 - #8 Vertical Bars E. F., #4 Ties @ 12" c/c



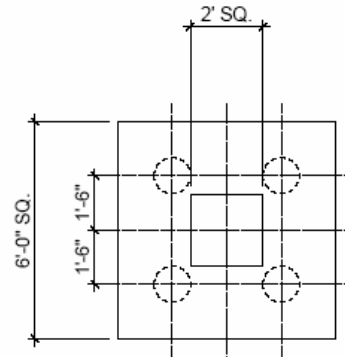
SAMPLE DESIGN 3: PIPERACK BRACED BAY FOUNDATION ON BATTERED PILES

PILE CAP DESIGN

Try 6' - 0" Square x 2' - 0" Thick Pile Cap
with 4 - Battered Piles

Pile Cap Weight

$$\begin{aligned} \text{Pier Weight} &= (2)(2)(4)(0.15) = 2.4 \text{ k} \\ \text{Cap Weight} &= (6)(6)(2)(0.15) = 10.8 \text{ k} \\ \text{Soil Weight} &= (62 - 22)(3)(0.1) = 9.6 \text{ k} \\ \text{Total} &= \underline{22.8 \text{ k}} \end{aligned}$$



Trial Pile Capacity

$$P_{\max} = 18 + 22.8 + 10.5 = 51.3 \text{ k}$$

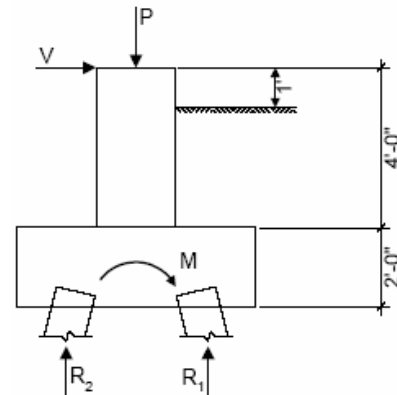
$$M = 7 \text{ k} \times 6.0' = 42.0 \text{ ft-k} \quad (S = 4(1.5)^2 / 1.5 = 6')$$

$$R_{1,2} = 51.3 / 4 \pm 42 / 6 = 12.8 \pm 7.0 = 19.8 \text{ k}, 5.8 \text{ k}$$

$$P_{\min} = 18 + 22.8 - 10.5 = 30.3 \text{ k}$$

$$R_{1,2} = 30.3 / 4 \pm 42 / 6 = 7.6 \pm 7.0 = 14.6 \text{ k}, 0.6 \text{ k}$$

Try to 10 T Pile



Computer Model

Note : Most "Hand" methods for battered piles ignore bending.

Joint Coordinates

Try :	S	X ₂	X ₃	X ₄
1	0.5	3.5	4.0	
2	1.0	4.0	5.0	
3	1.5	4.5	6.0	

Member Properties

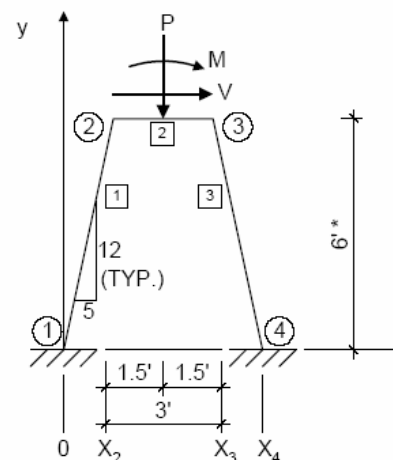
$$A_{1,3} = p / 4 \times 122 = 113 \text{ in.}^2$$

$$I_{1,3} = p / 64 \times 124 = 1,018 \text{ in.}^4$$

$$E_{1,3} = 1,500 \text{ ksi}$$

$$A_2 = 24 \times 36 = 864 \text{ in.}^2$$

$$I_2 = 1 / 12 \times 24 \times 36^3 = 41,472 \text{ in.}^4$$



* Dimension given by Geotechnical Consultant, per UBC or governing building code or determined from analysis, all with approval from Lead Structural Engineer.



SAMPLE DESIGN 3: PIPERACK BRACED BAY FOUNDATION ON BATTERED PILES

Design Loads

$$(P_{\max})_y = -51.3 / 2 = -25.65 \text{ k}$$

$$M_z - 42 / 2 = -21.0 \text{ ft-k}$$

$$V_x = 7 / 2 = 3.5 \text{ k}$$

$$(P_{\min})_y = -30.3 / 2 = -15.5 \text{ k}$$

$$M_z = -21.0 \text{ ft-k}$$

$$V_x = 3.5 \text{ k}$$

Analysis Results

S	Load	JOINT 1				JOINT 4			
		Mz	Rx	Ry	R	Mz	Rx	Ry	R
1	Pmax	7.0	0.7	5.8	5.8	7.0	2.8	19.8	20.0
	Pmin	7.0	1.1	0.6	1.3	7.0	2.4	14.6	14.8
2	Pmax	3.6	-0.4	5.8	5.8	3.4	3.9	19.8	20.2
	Pmin	3.5	0.5	0.6	0.8	3.5	3.0	14.6	14.9
3	Pmax	0.1	-1.4	5.8	6.0	0.1	4.9	19.8	20.4 — max
	Pmin	0.1	-.01	0.6	0.6	0.1	3.6	14.6	15.0

Notes :

1. Vertical Pile reactions (Ry) do not change with change in batter. (Same as those calculated under trial pile capacity.)
2. Bending Moment = 0 @ Joint 1 and Joint 5 for vertical loads alone.
3. When Bending is excessive, even @ 3 : 12 batter, consider Fixed Head condition.

Use 10 T Piles @ 3 : 12 Batter

$$(M_z = 0.1 \text{ ft-k} \longrightarrow \text{small})$$

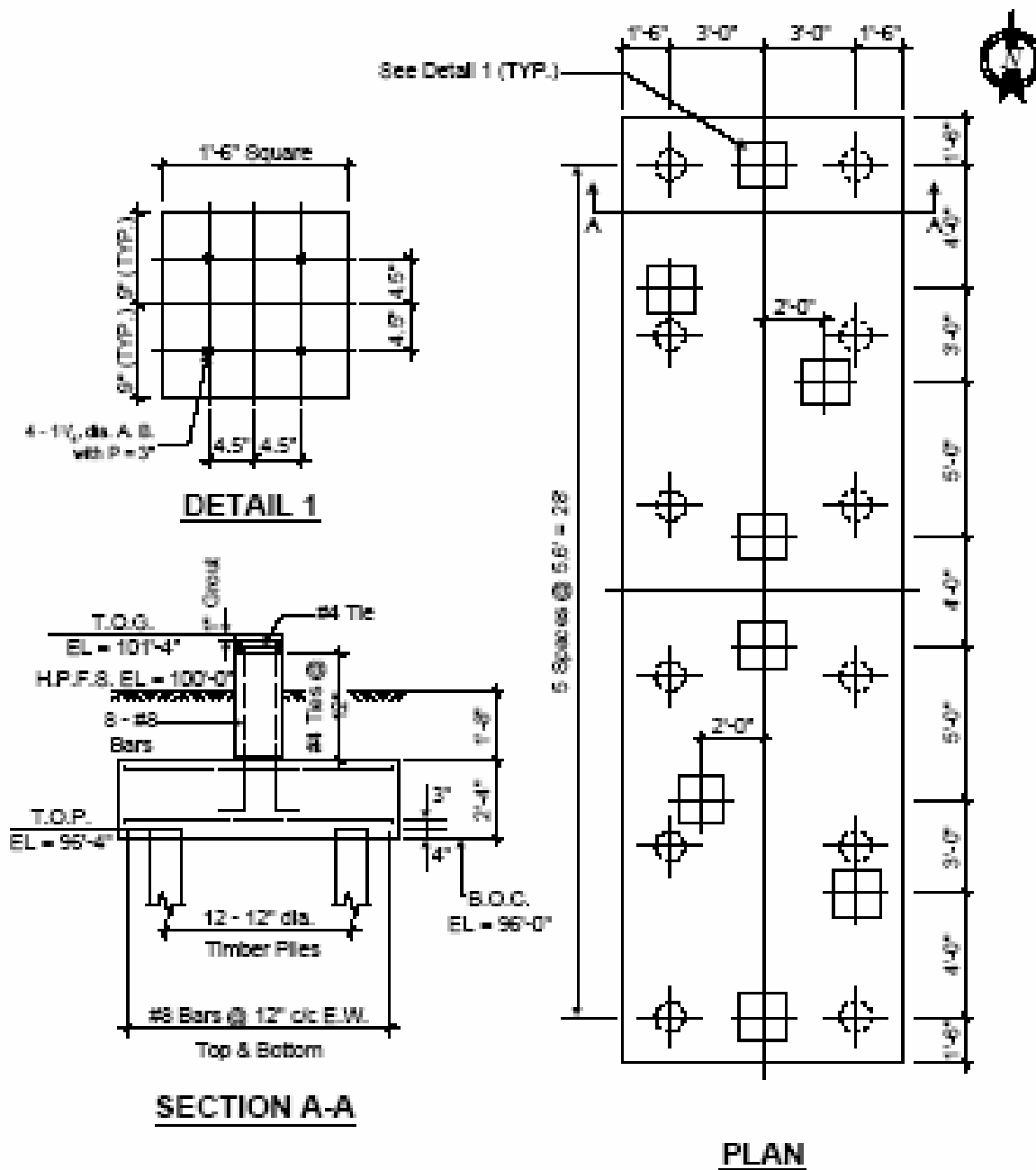
$$R_{\max} = 20.4 \text{ k} \approx 20 \text{ k} \longrightarrow \text{okay})$$

No uplift; however, uplift potential exists $\longrightarrow$ Use Standard Tension Connection

Reinforcement Design

Since piles are located within $D / 2$ of the Pier, shear, moment, and punching shear stresses in the Pile Caps are small; therefore, no calculations are required.

Use #5 Bars @ 12" c/c E.W., Top & Bottom

SAMPLE DESIGN 4 : COMBINED FOUNDATION ON PILES



SAMPLE DESIGN 4 : COMBINED FOUNDATION ON PILES

DESIGN DATA

References

ACI-318-89 for Concrete Design

Materials

Concrete : $f'_c = 3,000 \text{ psi}$, $\gamma_{\text{conc}} = 150 \text{ pcf}$
Rebar : $F_y = 40,000 \text{ psi}$
Piles : 12" dia. Timber Piles
Allowable Compression = 35 T
Allowable Tension = 15 T
Allowable Shear = 2 k
Minimum Embedment = 4"

$\frac{1}{3}$ Increase in Allowable Capacities for Load Combinations with Seismic Loads.

Anchor Bolts : 4' - 1 $\frac{1}{4}$ " dia. Anchor Bolts, A36 (4 Each Pier)

Soil : $\gamma_{\text{soil}} = 100 \text{ pcf}$

Design Loads

Vertical Loads are as defined on next sheet. $\Sigma = 500 \text{ k}$

Lateral Seismic Force @ Each Load = $0.2 \times (\text{Vertical Load in any direction})$

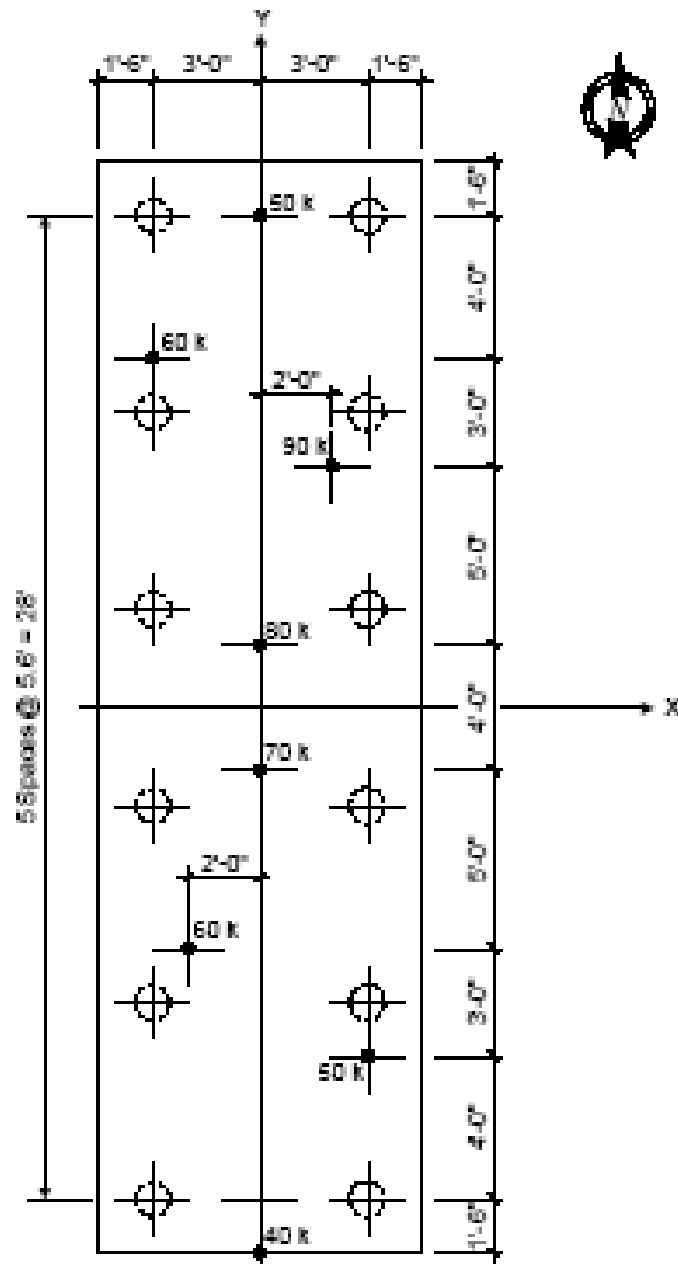
Moment Arm = 8' - 0" from Pile Cutoff

$M_{\text{seismic}} = 0.2 \times 500 \text{ k} \times 8' = 800 \text{ ft-k}$



SAMPLE DESIGN 4 : COMBINED FOUNDATION ON PILES

Design Model



PLAN



SAMPLE DESIGN 4 : COMBINED FOUNDATION ON PILES

PIER DESIGN

Note : For brevity, pier design including : pier dimensions, anchor bolt design, slide plate design, and reinforcement are not shown here. See Guideline 000-215-1225, Combined Foundations, for details of pier design.

Use 1' - 6" Square Pier, with 8 - #8 Vertical Bars, #4 Ties @ 12" c/c

PILE CAP DESIGN

Try 12 Piles

$$I_x = 4(2.8^2 + 8.4^2 + 14.0^2) = 1,098 \text{ ft}^2$$

$$I_y = 6 \times 2 \times 3^2 = 108 \text{ ft}^2$$

C. G. of Loads

P	(± Y)	=	M _x	(± X)	=	M _y
40k	(-14)	=	-560	(0)	=	0
50 k	(-10)	=	-500	(3)	=	150
60 k	(-7)	=	-420	(-2)	=	-120
70 k	(-2)	=	-140	(0)	=	0
80 k	(2)	=	160	(0)	=	0
90 k	(7)	=	630	(2)	=	180
60 k	(10)	=	600	(-3)	=	-180
<u>50 k</u>	<u>(14)</u>	=	<u>700</u>	<u>(0)</u>	=	<u>0</u>
500 k			470 ft-k			30 ft-k

$$\bar{Y} = 470 / 500 = 0.94'$$

$$\bar{X} = 30 / 500 = 0.06'$$

(Permanent Eccentricity)

Preliminary Foundation Weights

Note : A 2' - 4" Pile Cap is assumed. Additional weight may be neglected.

Pier Weights =	(1.5 ² x 3 x 0.15) k / Pier x 8 Piers =	8.0 k
Pile Footing =	9 ft x 31 ft x 2 ft x 0.15 =	83.7 k
Soil =	9 ft x 31 ft x 2 ft x 0.10 =	<u>55.8 k</u>
	Total =	<u>147.5 k</u>



SAMPLE DESIGN 4 : COMBINED FOUNDATION ON PILES

Pile Loads

Operating

Rows I, II :

$$P / N \pm \Sigma M_x c / I = (500 + 147.5) / 12 \pm 470 \times 14 / 1098 \pm 30 \times 3 / 108 = 61 \text{ k}, 47 \text{ k} < 70 \text{ k (okay)}$$

Long-term settlement will be negligible with permanent eccentricity.

Operating + Seismic

Seismic (North - South), Rows I, II :

$$P / N \pm M_x(y) / I_x \pm M_y(x) / I_y \pm M_{\text{seismic}}(y) / I_x = 61 \pm 800 \times 14 / 1098 = 71 \text{ k}, 51 \text{ k} < 1.33 \times 70 \text{ k (okay)}$$

Seismic (East - West), Rows I, II :

$$P / N \pm M_x(y) / I_x \pm M_y(x) / I_y \pm M_{\text{seismic}}(x) / I_y = 47 \pm 800 \times 3 / 108 = 69 \text{ k}, 25 \text{ k} < 1.33 \times 70 \text{ k (okay)}$$

$$\text{Max Pile Reaction} = 61 + \sqrt{10.2^2 + 22.2^2} = 85 \text{ k} < 1.33 \times 70 \text{ k (okay)}$$

Use 12 - 12" dia. Timber Piles

Ultimate Reactions due to Eccentricity & Seismic (North - South)

$$U = 1.05D + 1.40E \text{ (ACI 318-89, Section 9.2, 3)}$$

$$M_x = 1.4(800) + 1.05(870) = 1613.5 \text{ ft-k}$$

$$M(y) / I = 1613.5 / 1098 = 1.47y$$

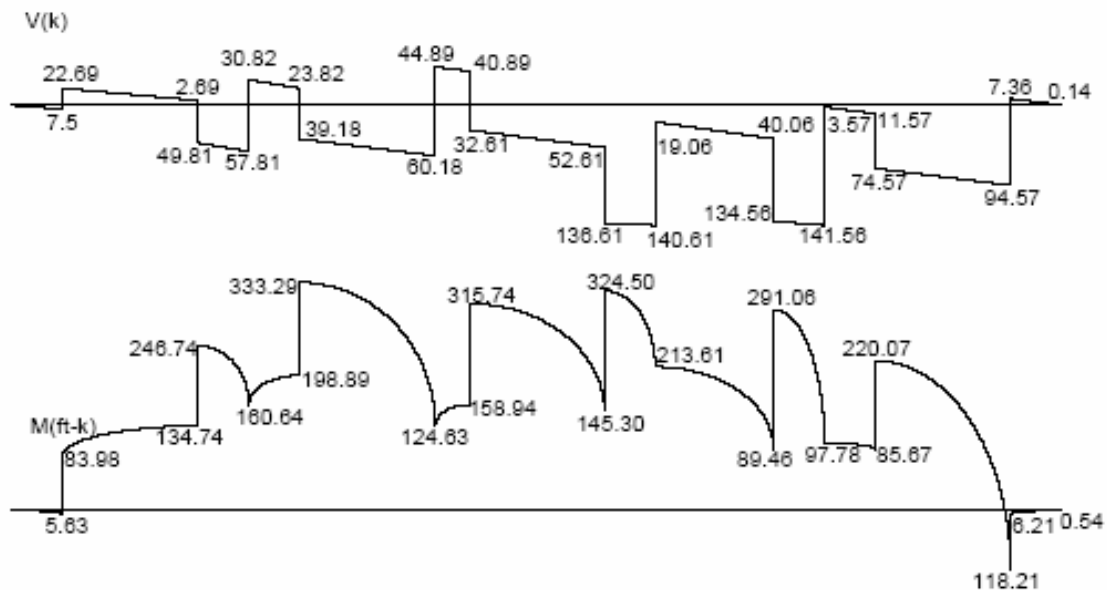
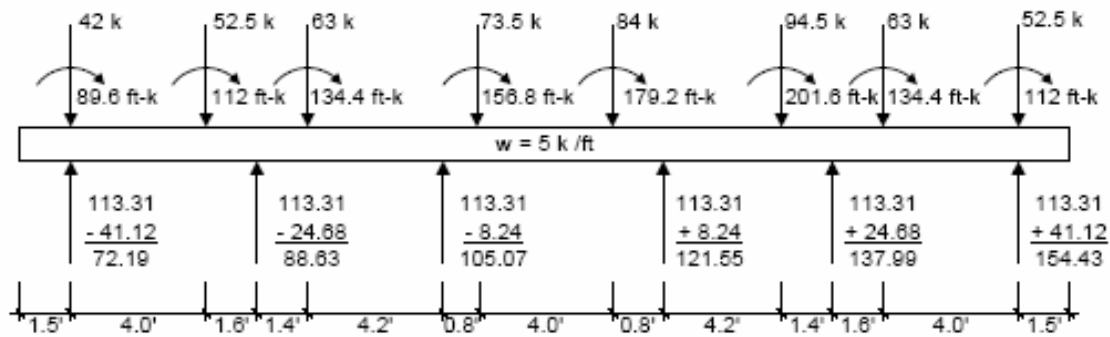
#1 =	1.47 x 14.0 =	±20.56 k / per Pile	(x 2 Piles = 41.12 k / Row)
#2 =	1.47 x 8.4 =	±12.34k / per Pile	(x 2 Piles = 24.68 k / Row)
#3 =	1.47 x 2.8 =	±4.12 k / per Pile	(x 2 Piles = 8.24 k / Row)

Uniform Loads

$$w = 1.05 \times 147.5 / 31 = 5 \text{ k / ft}$$



SAMPLE DESIGN 4 : COMBINED FOUNDATION ON PILES



SAMPLE DESIGN 4 : COMBINED FOUNDATION ON PILES**Critical Transverse Strips**

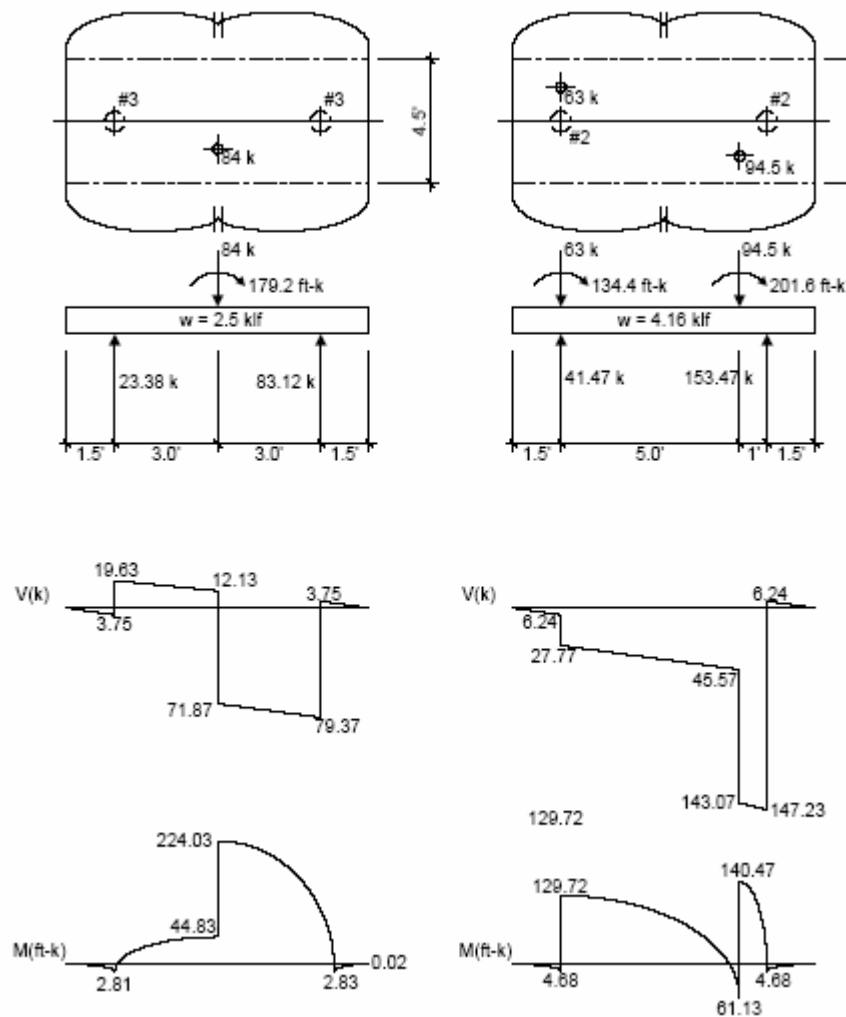
Based on distance between loads

$$\text{Strip Width} = (5 + 4) / 2 = 4.5'$$

$$= 5 / 2 + 3 + 4 / 2 = 7.5'$$

$$w = (1.05 \times 147.5 \times 4.5) / (31 \text{ ft} \times 9 \text{ ft}) = 2.5 \text{ klf}$$

$$= (1.05 \times 147.5 \times 7.5) / (31 \text{ ft} \times 9 \text{ ft}) = 4.16 \text{ klf}$$

Maximum Moment**Maximum Shear****Check Punching Shear**



SAMPLE DESIGN 4 : COMBINED FOUNDATION ON PILES

Check "Corner" Pile

For 2' - 4" Thick Footing $\rightarrow d = 28 - 3 - 4 = 21"$

$$b_o = 4(b + d) = 4(10.63 + 21) = 127"$$

OR

$$b_o = 2(21 / 2 + 10.63 / 2 + 18) = 68" \leftarrow \text{Use}$$

$$\phi V_n = \phi 4 \sqrt{f'_c} b_o d$$

$$= 0.85 \times 4 \times \sqrt{3,000} \times 68 \times 21 / 1,000$$

$$= 266 \text{ k} > 35 \text{ T} \times 2 \text{ (T / k)} \times 1.4 = 98 \text{ k}$$

000

Reinforcement Design

Top Steel

$$d = 21", b = 12"$$

$$M_{u, \text{MAX-LONG}} = 333.29 / 9 = 37.0 \text{ ft-k / ft}$$

$$M_{u, \text{MAX-TRANS}} = 224.03 / 4.5 = 49.8 \text{ ft-k / ft} \leftarrow \text{Controls}$$

$$F = 12 \times 21^2 / 12,000 = 0.6615$$

$$K_u = 49.8 / 0.6615 = 75.28 \rightarrow a_u = 4.43$$

$$A_{s, \text{Req}} = 49.8 / (4.43 \times 21) = 0.54 \text{ in.}^2 / \text{ft}$$

$$\frac{4}{3} A_s = 0.71 \text{ in.}^2 / \text{ft} \leftarrow \text{Controls}$$

$$A_{s, \text{min}} = 0.0033(12)(21) = 0.83 \text{ in.}^2 / \text{ft}$$

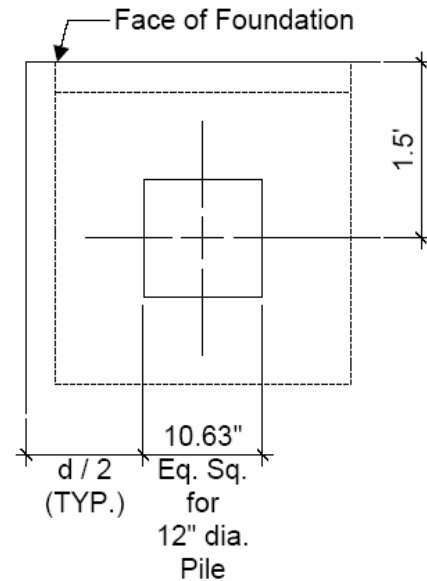
Use #8 Bar @ 12" E.W., Top & Bottom ($A_s = 0.79 \text{ in.}^2 / \text{ft}$)

Beam Shear

$$V_{u, \text{MAX-LONG}} = 141.56 / 9 = 15.7 \text{ k / ft}$$

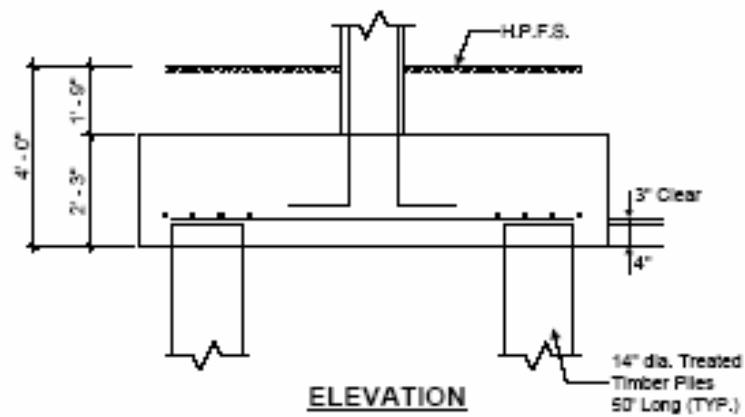
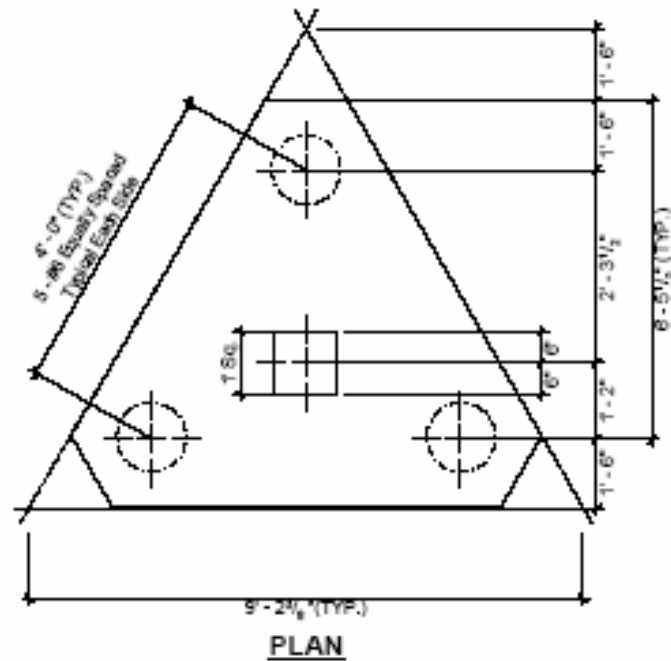
$$V_{u, \text{MAX-TRANS}} = 147.23 / 7.5 = 19.63 \text{ k / ft} \leftarrow \text{Controls}$$

$$V_u = V_u / \phi b d = 19.63 / (0.85 \times 12 \times 21) = 92 \text{ psi} < 2 \sqrt{3,000} = 110 \text{ psi (okay)}$$





SAMPLE DESIGN 5 : THREE PILE FOUNDATION FOR T - SUPPORT





SAMPLE DESIGN 5 : THREE PILE FOUNDATION FOR T - SUPPORT

DESIGN DATA

References

ACI-318-89 for Concrete Design

Materials

Concrete : $f'_c = 3,000$ psi, $\gamma_{conc} = 150$ pcf
Rebar : ASTM A-615 Grade 60
Soil : Passive Pressure = 200 psf / ft of depth
Bottom of Pile Cap @ 4' - 0" Below H.P.S.F.
Piles : 50' Long Treated Timber per ASTM D25-73
Compression Capacity = 30 T
Uplift Capacity = 15 T
Lateral Capacity = 2 k
Butt Diameter = 14"
Minimum Pile Spacing = 3' - 0"
Minimum Edge Distance = 1' - 6"
Minimum Embedment = 4"

Allowable Increase for Piles and Soil :

$$\text{Wind or Seismic} = \frac{1}{3}$$

$$\text{Test} = \frac{1}{5}$$

Design Loads

Note : For brevity, design loads and load combinations are not shown here. See Guideline 000-215-1250, Pipe Supports, for required design load information. Assume that analysis provides the following loads at the top of the pile cap :

Dead Load = 12.5 k

Live Load = 33.0 k

Test Load = None

Lateral Seismic Load = 7.5 k @ 8' - 0" above bottom of pile cap in any direction

T - SUPPORT DESIGN

Note : For brevity, the design of the concrete T - support on the three pile foundation is omitted. See Guideline 000-215-1250, Pipe Supports, for details of design.



SAMPLE DESIGN 5 : THREE PILE FOUNDATION FOR T - SUPPORT

PILE CAP DESIGN

Trial Size

Proportions are set by equilateral triangle
and "a" is a function of the load.

Solve for "a" (Estimated Wt. of Soil + Cap = 20 k)

$$P_{\text{oper}} \approx \frac{125 + 33 + 20}{3} = 59.3 \text{ k / Pile} \leq 60 \text{ k / Pile Allow. (okay)}$$

$$P_{\text{oper} + \text{seismic}} \approx 60 \text{ k / Pile} \pm \frac{7.5 \text{ k} \times 8 \text{ ft}}{0.866 \times a} \leq 60 \text{ k / Pile (x 1.33)}$$

$$\rightarrow a \geq 3.5' > 3.0' \text{ Min. (okay)}$$

Use 4' - 0" Pile Spacing

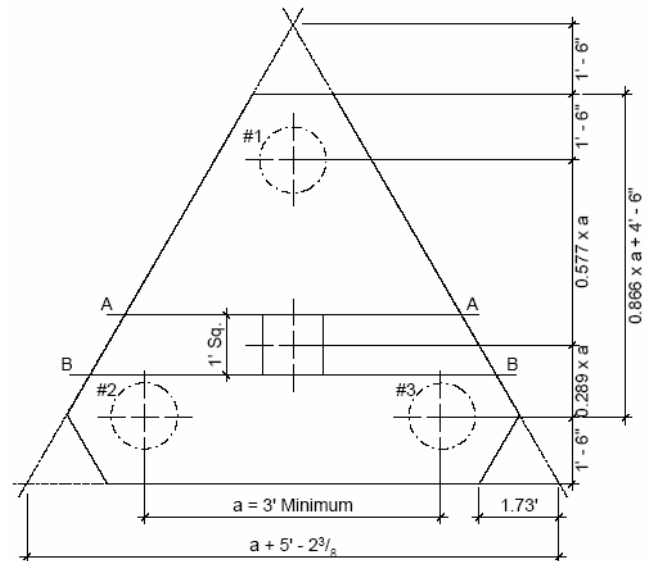
Pile Reactions

$$P_{\#1, \text{oper} + \text{seismic}} = 60 \text{ k} \pm \frac{7.5 \text{ k} \times 8 \text{ ft}}{0.866 \times 4 \text{ ft}} = 77.3 \text{ k, } 42.7 \text{ k}$$

$$P_{\#2} = P_{\#3, \text{oper} + \text{seismic}} = 60 \text{ k} \pm \frac{7.5 \times 8 \text{ ft}}{2 \times 0.866 \times 4 \text{ ft}} = 68.7 \text{ k, } 51.3 \text{ k}$$

$$P_{\text{max, oper}} = 60 \text{ k / Pile} \leftarrow \text{Governs}$$

$$P_{\text{max, oper} + \text{seismic}} = \frac{77.3 \text{ k / Pile}}{1.33} = 58.1 \text{ k / Pile}$$



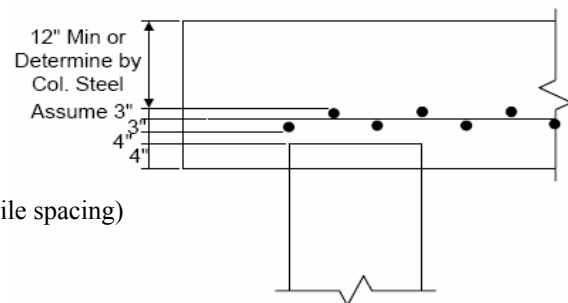
Shear in Cap

Determine "d"

Critical Sections for Shear :

1. "d" from face of column; ACI 11.1.3.1 (usually governs)
2. "d" from face of equivalent square pile (may govern for large pile spacing)
3. "d / 2" from periphery of column or piles; ACI 11.1.3.2

$$\text{Equivalent Square} = \sqrt{\pi(14'')^2} = 12.4'' \approx 12''$$



Cap Thickness

$$t_{\text{min}} = 1' - 10''$$

$$\text{Try } t = 2' - 3''; d = 18.5''$$

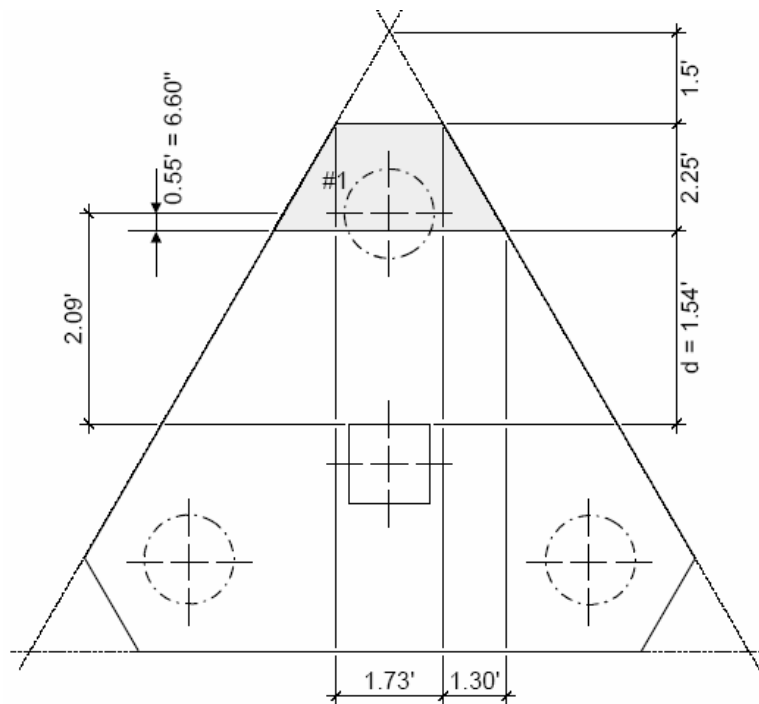


SAMPLE DESIGN 5 : THREE PILE FOUNDATION FOR T - SUPPORT

$$\text{Cap Area} = 8.39'(4.85') - 3(1.5')(0.865') = 36.8 \text{ ft}^2$$

$$\text{Wt. Cap} + \text{Soil} = [(0.10 \text{ kcf})1.75' + (0.15)(2.25)]36.8 \text{ ft}^2 = 18.9 \text{ k} \leq 20 \text{ k Assumed (okay)}$$

Check Shear @ d = 22" from A - A



$$\text{Factored Shear due to Soil} + \text{Cap} = 0.513(1.73' + 1.30') 2.25' \times 1.4 = 4.9 \text{ k}$$

Center of pile is located less than $\frac{d_{pile}}{2}$ from the section and therefore the reaction does not produce full shear on the section.

$$\text{Factored Effective Pile Reaction} = \frac{13.6}{4} \frac{[(125k + 18.9k)1.4 + 33k(1.7)]}{3} = 83.4 \text{ k}$$

$$\text{Net Factored Shear} = V_u = 83.4 - 4.9 = 78.5 \text{ k}$$

$$\bar{V}_u = \frac{V_u}{\phi b d} = \frac{78.5}{0.85(51.96)(18.5)} = 96 \text{ psi} < 2 \sqrt{f'_c} = 110 \text{ psi (okay)}$$

Check Shear @ d = 18.5" from B - B

This section occurs 8" from edge of cap and is not critical. (By Inspection)

SAMPLE DESIGN 5 : THREE PILE FOUNDATION FOR T - SUPPORT

Check Shear @ d= 18.5" from Piles #2 and #3

This section is located on the opposite side of the column and therefore does not represent a possible plane of failure with respect to Piles #2 and #3.

Check Shear @ d = 18.5" from Pile #1

This section is not critical as it falls within the column.

Punching Shear @ d/2 from Periphery

Column

$$V_u = 1.4(125 \text{ k}) + 1.7(33) - 1.4(0.513)(1 + 1.83')^2 = 225.3 \text{ k}$$

$$\bar{V}_u = \frac{225.3 \text{ k}}{0.85(4)(30.5)(18.5)} = 117.4 \text{ psi} < 4\sqrt{f'_c} \text{ (okay)}$$

Piles

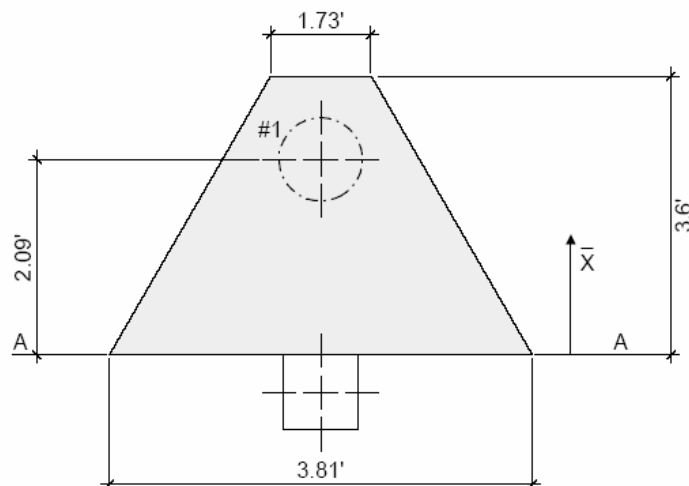
$$V_u = 85.9 - 1.4(0.513 \text{ ksf})\left(\frac{\pi}{4}\right)\left(\frac{14 + 22''}{12}\right)^2 = 80.8 \text{ k}$$

$$\bar{V}_u = \frac{80.8 \text{ k}}{0.85(32.5)(\pi)(18.5)} = 50.3 \text{ psi (okay) } 80.8 \text{ k}$$

Bending in Cap

Governing section occurs at A - A face of column. Pile reaction is considered as a point load.

$$\bar{X} = \frac{3.6}{3} \left[1 + \frac{1}{1 + \frac{3.81}{1.73}} \right] = 1.57'$$





SAMPLE DESIGN 5 : THREE PILE FOUNDATION FOR T - SUPPORT

$$A = 3.6 \left(\frac{1.73 + 3.81}{2} \right) = 9.97 \text{ ft}^2$$

$$P_{w, \text{cap} + \text{soil}} = 0.513(9.97 \text{ ft}^2)1.4 = 7.16 \text{ k}$$

$$M_{\text{Col. Face}} = 86.5 \text{ k}(2.09') - 7.16 \text{ k}(1.57') = 169.5 \text{ ft-k}$$

$$K_u = \frac{12,000 M_u}{b d^2} = \frac{169.5(12,000)}{45.7(18.5)^2} = 130.0$$

$$a_u = 4.37$$

$$A_{s, \text{Req'd}} = \frac{169.5}{4.37(18.5'')} = 2.10 \text{ in.}^2$$

$$\frac{4}{3} A_s = \frac{4}{3} (2.10) = 2.80 \text{ in.}^2$$

$$A_{s, \text{Min}} = 0.003(18.5'')(45.72'') = 2.82 \text{ in.}^2$$

OR

$$A_{s, \text{Min}} = \frac{2.82}{\cos 30^\circ} = 3.25 \text{ in.}^2 @ 60^\circ \text{ to A - A}$$

Use 10 - #6 bars (5 Each band) ($A_s = 4.40 \text{ in.}^2$)

Development Length

$$l_{db} = 19 \left(\frac{3.25}{4.40} \right) = 14''$$

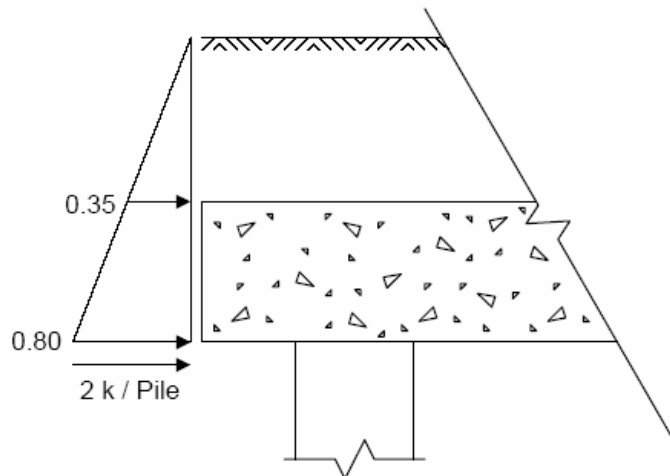
$$l_{db, \text{Provided}} = 16.9'' \text{ (okay)}$$

Lateral Resistance

Resistance Provided

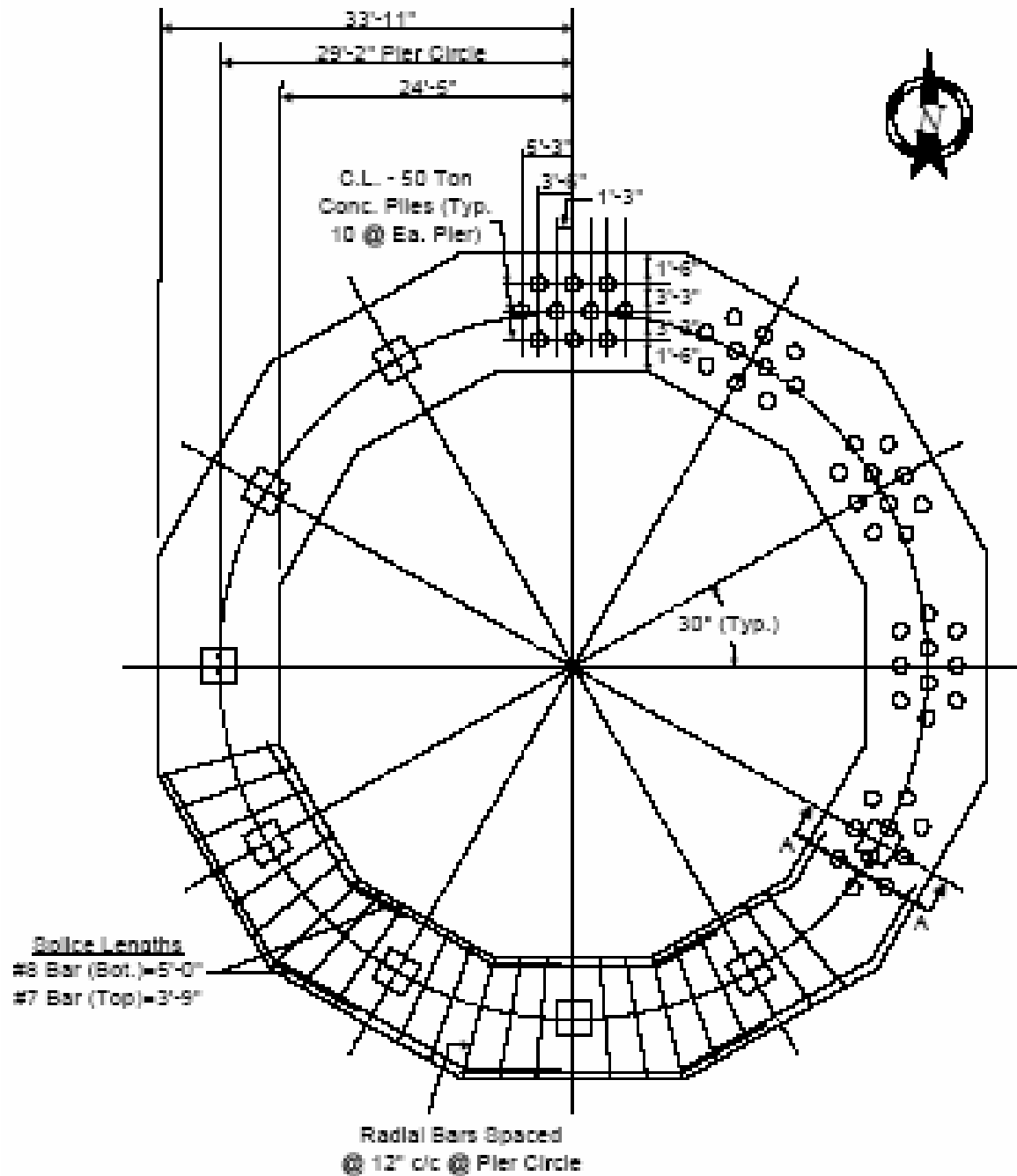
$$= \frac{0.8 + 0.35}{2} \times 2.25'(6.46') + 3(2 \text{ k})$$

$$= 14.4 \text{ k} > 7.5 \text{ k (okay)}$$





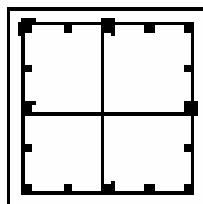
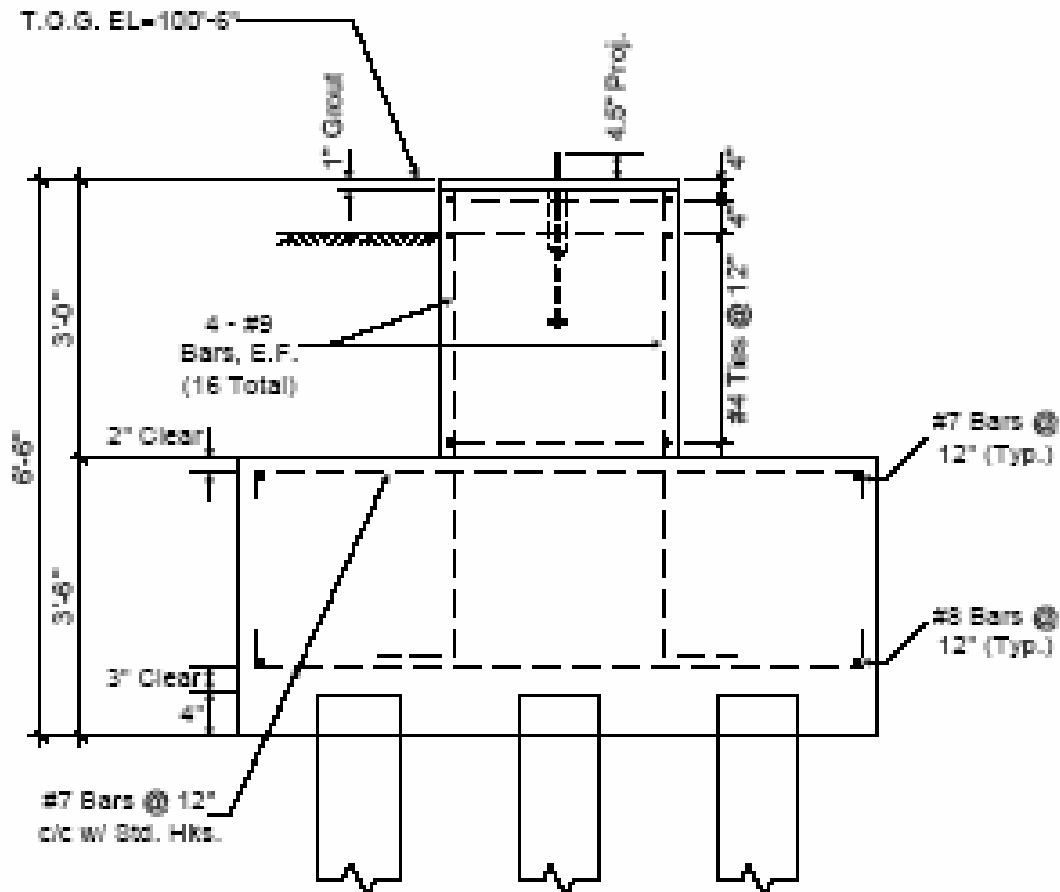
SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES



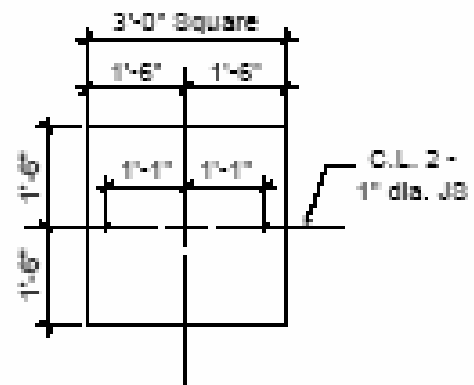
PLAN



SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES



TIE DETAIL





SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES

REFERENCES

ACI-318-89
Vessel Data Sheet, Load
Calculations & Vendor Drawings

DESIGN DATA

$f'_c = 3,000$ psi

$F_y = 60,000$ psi

Soil Weight = 120 pcf

Passive Soil Resistance = 200 psf / ft of depth

Concrete Piles : Allow. Capacities

Compression Capacity = 50 T

Uplift Capacity = 23 T

Lateral Resistance = 2 k

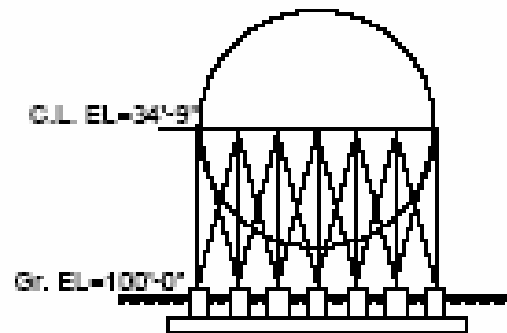
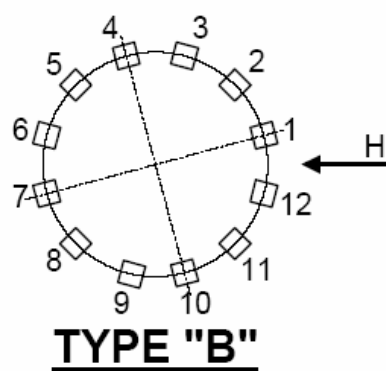
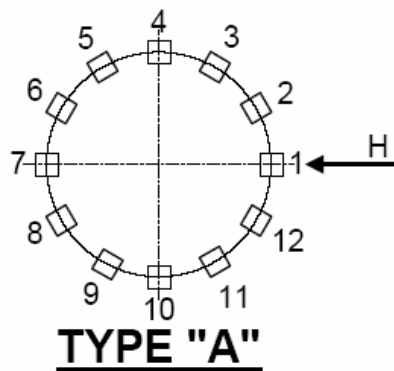
No reduction for Pile Group necessary.

Allowable increase for Piles and Passive Soil Pressure: Wind or Seismic = $\frac{1}{3}$ Increase

Test = $\frac{1}{5}$ Increase

LOADINGS

All loads shown on Sheet 2 are provided by the vendor and will be used for the design. All vendor supplied loads, however, should be checked by the engineer to ensure their accuracy. Loads are given for the Type "A" arrangement only. Loads for the Type "B" arrangement should be provided and investigated but will not be covered in this design.



Data from Vendor

SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES

TYPE "A" ARRANGEMENT (ALL FORCES IN KIPS AND MOMENTS IN FT-K.)

LOADS AT TOP OF PIERS													
Pier No.	VERTICAL							HORIZONTAL*			FIXED-END MOMENT		
	Emp.	Oper.	Test	Live	Seis. Emp.	Seis. Oper.	Wind	Seis. Emp.	Seis. Oper.	Wind	Seis. Emp.	Seis. Oper.	Wind
1.0	36.5	312.0	573.0	5.0	-17.1	-146.6	-11.1	2.0	16.6	1.4	0.0	0.0	0.0
2.0	36.5	312.0	573.0	5.0	-14.8	-126.5	-9.6	7.3	62.4	4.8	0.0	0.0	0.0
3.0	36.5	312.0	573.0	5.0	-8.7	-73.0	-5.6	13.6	116.0	8.9	0.0	0.0	0.0
4.0	36.5	312.0	573.0	5.0	0.0	0.0	0.0	13.6	116.0	8.9	0.0	0.0	0.0
5.0	36.5	312.0	573.0	5.0	8.7	73.0	5.6	7.3	62.4	4.8	0.0	0.0	0.0
6.0	36.5	312.0	573.0	5.0	14.8	126.5	9.6	1.0	8.3	0.7	0.0	0.0	0.0
7.0	36.5	312.0	573.0	5.0	17.1	146.6	11.1	0.0	0.0	0.0	0.0	0.0	0.0
8.0	36.5	312.0	573.0	5.0	14.8	126.5	9.6	1.0	8.3	0.7	0.0	0.0	0.0
9.0	36.5	312.0	573.0	5.0	8.7	73.0	5.6	7.3	62.4	4.8	0.0	0.0	0.0
10.0	36.5	312.0	573.0	5.0	0.0	0.0	0.0	13.6	116.0	8.9	0.0	0.0	0.0
11.0	36.5	312.0	573.0	5.0	-8.7	-730.0	-5.6	13.6	116.0	8.9	0.0	0.0	0.0
12.0	36.5	312.0	573.0	5.0	-14.8	-126.5	-9.6	7.3	62.4	4.8	0.0	0.0	0.0
Σ	438.0	3,744.0	6,876.0	60.0	0.0	0.0	0.0	87.6	746.8	57.6	0.0	0.0	0.0

Notes : * The horizontal component at each pier is parallel with the assumed direction of wind or seismic.

Base Plate details not shown.

FOUNDATION DESIGN

Case A : Individual Column Foundations

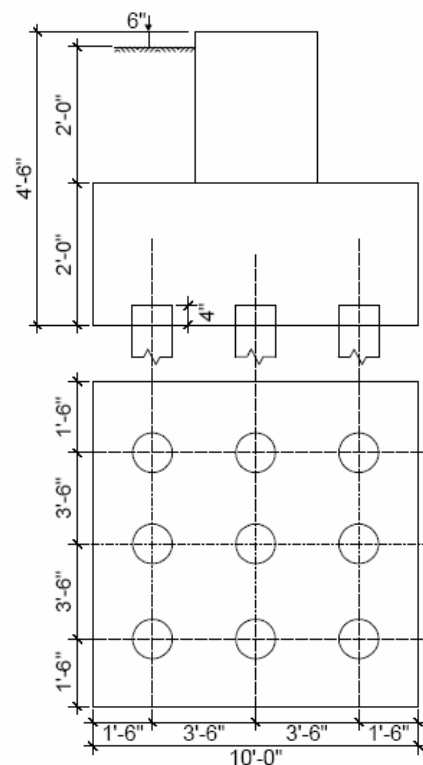
1. Test + Live Load

Vertical Load on Pier = $573.0 + 5.0 = 578.0$ k

$$\begin{aligned}
 \text{Wt. of Pier} &= (3 \times 3)(2.5)(0.15) = 3.4 \text{ k} \\
 \text{Wt. of Footing} &= (10 \times 10)(2.0)(0.15) = 30.0 \text{ k} \\
 \text{Wt. of Soil} &= [(10 \times 10) - (3 \times 3)](2.0)(0.12) = 21.8 \text{ k} \\
 \text{Total} &= 55.2 \text{ k}
 \end{aligned}$$

Total Load per Foundation

$$P_{\text{total}} = 578.0 + 55.2 = 633.2 \text{ k}$$





SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES

$$\text{Load per Pile} = \frac{633.2}{9} = 70.4 \text{ k} < 100 \text{ k} \times 1.2 \text{ (okay)}$$

2. Oper. + Seismic

$$H = 116 \text{ k}, P = 312 \text{ k}$$

The number of piles required to resist vertical loads will be governed by the test condition.

Check resistance to horizontal force.

PASSIVE SOIL RESISTANCE

$$\text{Horizontal Force taken by Vertical Piles} = 9 \times 2 \times \frac{4}{3} = 24 \text{ k}$$

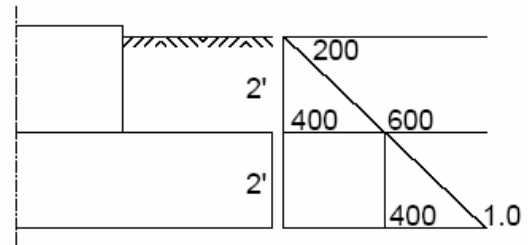
$$\text{Therefore } 116 \text{ k} - 24 \text{ k} = 92 \text{ k}$$

Horizontal force must be resisted by passive pressure on footing (including top 2' - area paved).

Passive Soil Pressure

$$(3' \times 2')(0.20 + \frac{0.40}{2}) + (10' \times 2')(0.60 + \frac{0.40}{2}) = 18.4 \text{ k (not good)}$$

To develop the required resistance would mean an excessively large footing. The friction at the bottom of the footing can not be considered because the soil may settle away from the bottom of the footing.



Try Battered Piles

Assume 3 : 12 batter using both tension and compression piles with :

$$\text{Maximum Allowable Horizontal Resistance per Pile} = 46 \text{ k} \times \frac{1}{4.12} \times \frac{4}{3} = 14.9 \text{ k},$$

based on tension capacity.

$$\text{Total Number of Battered Piles Required} = \frac{116 \text{ k}}{14.9 \text{ k}} = 8 \text{ Piles per Foundation}$$

Due to the number of battered piles required and the close proximity of the adjacent footings, pile interference can not be avoided.

Therefore, Use a Continuous Ring Foundation

SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES

Case B : Continuous Ring Foundation with Piles

1. Test + Live Load

$$\text{Total Vertical Load} = 6,876 + 60 = 6,936 \text{ k}$$

Area of a Regular Polygon

$$A = N(\tan\phi)(R_1^2 - R_2^2); \text{ where } N = \text{Number of Sides} = 12$$

$$\phi = \frac{180^\circ}{N} = 15^\circ$$

$$A = 12(\tan 15^\circ)(33.92^2 - 24.42^2) = 1,782 \text{ ft}^2$$

Assume 3' - 6" Foundation Thickness

$$\begin{aligned} \text{Wt. of Piers} &= 12(3 \times 3 \times 3.0)(0.15) = 48.5 \\ \text{Wt. of Ring} &= 1,782 \times 3.5 \times 0.15 = 935.6 \\ \text{Wt. of Soil} &= (1,782 - 3 \times 3 \times 12)(2.5)(0.12) = 502.3 \\ \text{Total} &= 1,486.4 \text{ k} \end{aligned}$$

$$\text{Total Load on Piles} = 6,936 + 1,486.4 = 8,422 \text{ k}$$

$$\text{Load @ Point under each Pier} = \frac{8,422}{12} = 702 \text{ k}$$

$$\text{Number of Piles Required under Pier} = \frac{702 \text{ k}}{100 \times 1.2} = 5.8 \text{ Piles}$$

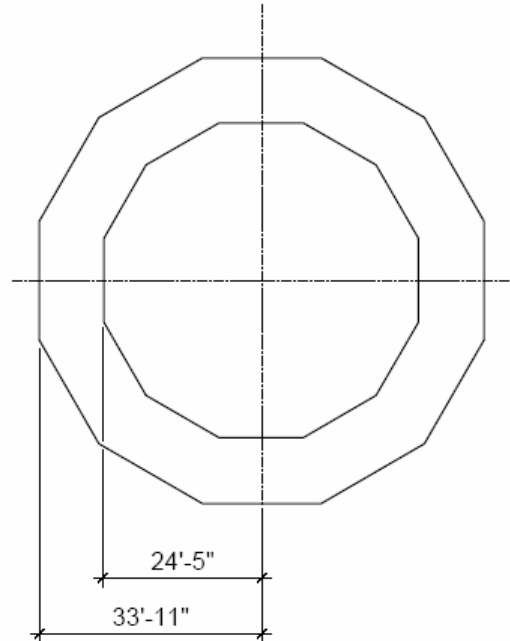
Use 10 Piles under each Pier (120 Total)

2. Oper. + Seismic

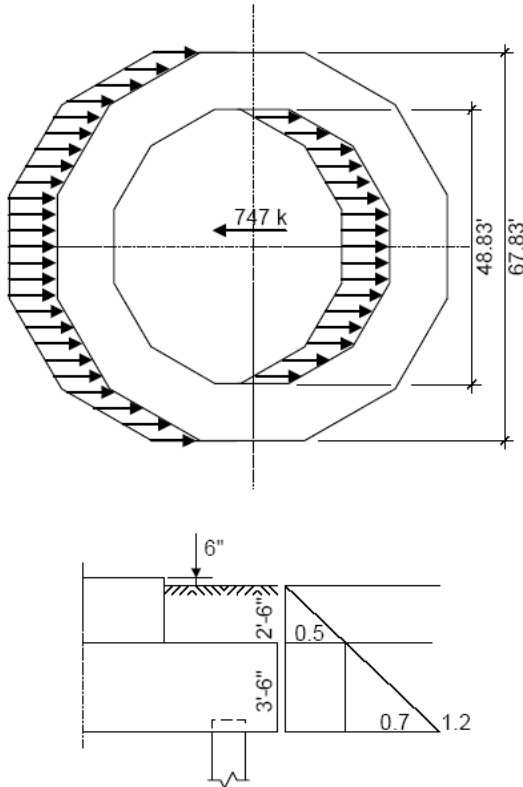
$$\text{Horizontal Force resisted by Piles} = 120 \times 2 \times \frac{4}{3} = 320 \text{ k}$$

Therefore $747 \text{ k} - 320 \text{ k} = 427 \text{ k}$ must be resisted by passive pressure on the foundation.

Note : e engineer should exercise caution in using passive soil pressure to resist lateral seismic loads especially when dealing with a cohesionless soil and / or backfill because of its tendency to produce a quick condition under repeated shocks.



SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES



Total Effective Length for Passive Resistance

$$l = 67.83 + 48.83 = 116.66'$$

Passive Resistance

$$(0.50 \times 3.5 + \frac{1}{2} (0.70)(3.5)) \times 116.66 \times \frac{4}{3} = 462.7 \text{ k} > 427 \text{ k (okay)}$$

Check Vertical Load on Piles (Pier Circle = 58' - 4")

$$M @ \text{Top of Piles} = 747 \text{ k} \times 40.8' = 30,478 \text{ ft-k}$$

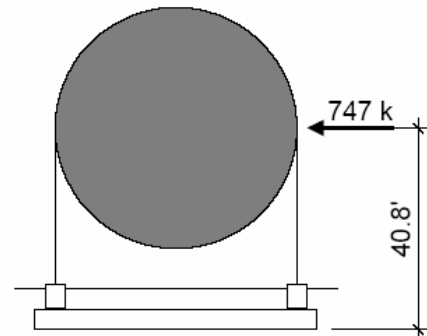
Wt. Piers =	$12(3 \times 3 \times 3.0)(0.15) =$	48.6 k
Wt. Ring =	$1,782(3.5)(0.15) =$	935.6 k
Wt. Soil =	$(1,782 - 3 \times 3 \times 12)(2.5)(0.12) =$	502.2 k
	Total =	1,486.4 k

$$I_{xx} = \frac{Nd^2}{8} = \frac{12(58.33)^2}{8} = 5,104$$

$$\text{Total Vertical Load} = 3,744 + 1,486.4 = 5,230.4 \text{ k}$$

$$P = \frac{5,230.4}{12} \pm \frac{30,478(29.17)}{5,104} = 610 \text{ k} < 10(100) \frac{4}{3} \text{ (okay)}$$

262 k (No Uplift)



Note : Due to large diameter of pile circle and the compactness of each pile group, assume the load to be equally distributed to each pile within a group with negligible error.

SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES

PIER DESIGN

Note : For brevity, pier design including pier dimensions, anchor bolt design, and reinforcement are not shown. See Guideline 000-215-1223 "Sphere Foundations" for details.

Use 16 - #9 Bars, 4 E.F. with #4 Ties

FOOTING DESIGN

Check Development

For #9 Bars :

$$l_{dh} = \frac{1,220d_p}{\sqrt{f'_c}} = \frac{1,200(1.128)}{\sqrt{3,000}} = 25"$$

$$T_{req'd} = 3" + 4"<emb.> + 2(1.0) + 25" = 34" < 42" \text{ (okay)}$$

Footing Reinforcing

$$d = 42" - 3" - 4" - 1.5"(1") = 33.5" \\ = 2.79'$$

Assume Critical Shear Plane
 @ "d" from Face of Pedestal

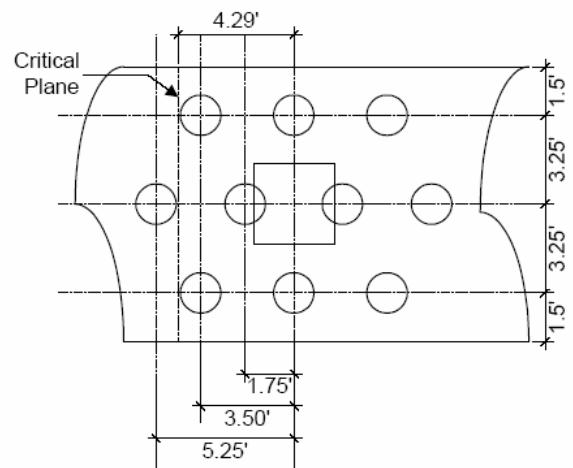
Max Load per Pile (Refer to Pg.4)

$$1.4(693 \text{ k}) , 10 \text{ Piles} = 97 \text{ k (Test + LL)}$$

Neglect Wt. of Soil + Conc. - Consv.

$$V_u = \frac{97 \times 1,000}{0.85(9.5 \times 12)(33.5)} = 29.9 \text{ psi} < 2 \cdot 97 \sqrt{f'_c}$$

$$= 109.5 \text{ psi (okay)}$$





SAMPLE DESIGN 6: SPHERE FOUNDATION ON PILES

Radial Reinforcing (Test + LL)

Moment @ Face of Ped. = $97 \text{ k} (3)(3.25 - 1.5) = 509 \text{ ft-k} \div 8' = 64 \text{ ft-k} / \text{ft} \sim$

$$F = \frac{bd^2}{12,000} = \frac{12(33.5)^2}{12,000} = 1.12$$

$$K_u = \frac{M_u}{F} = \frac{64}{1.12} = 57.1 \rightarrow a_u = 4.44$$

$$A_s = \frac{M_u}{a_u \times d} = \frac{64}{4.44(33.5)} = 0.43 \text{ in}^2 / \text{ft}$$

$$\frac{4}{3} A_s = 0.57 \text{ in}^2 / \text{ft} \leftarrow \text{Governs}$$

$$A_{s,\min} = 0.0033(12)(33.5) = 1.33 \text{ in}^2 / \text{ft}$$

Use #7 Bars @ 12" c/c Top & Bottom with Hook Bars to Develop ($A_s = 0.60 \text{ in}^2 / \text{ft}$)

Bottom Longitudinal Reinforcing (Test + LL)

Moment @ Face of Ped. = $97 \text{ k}(3.75) + 2(97 \text{ k})(2.0) + 97 \text{ k}(0.25) = 776 \text{ ft-k} , 95' = 81.7 \text{ ft-k} / \text{ft}$

$$K_u = \frac{M_u}{F} = \frac{81.7}{1.12} = 72.9 \rightarrow a_u = 4.43$$

$$A_s = \frac{M_u}{a_u \times d} = \frac{81.7}{4.44(33.5)} = 0.55 \text{ in}^2 / \text{ft}$$

$$\frac{4}{3} A_s = 0.73 \text{ in}^2 / \text{ft} \leftarrow \text{Governs}$$

$$A_{s,\min} = 1.33 \text{ in}^2 / \text{ft}$$

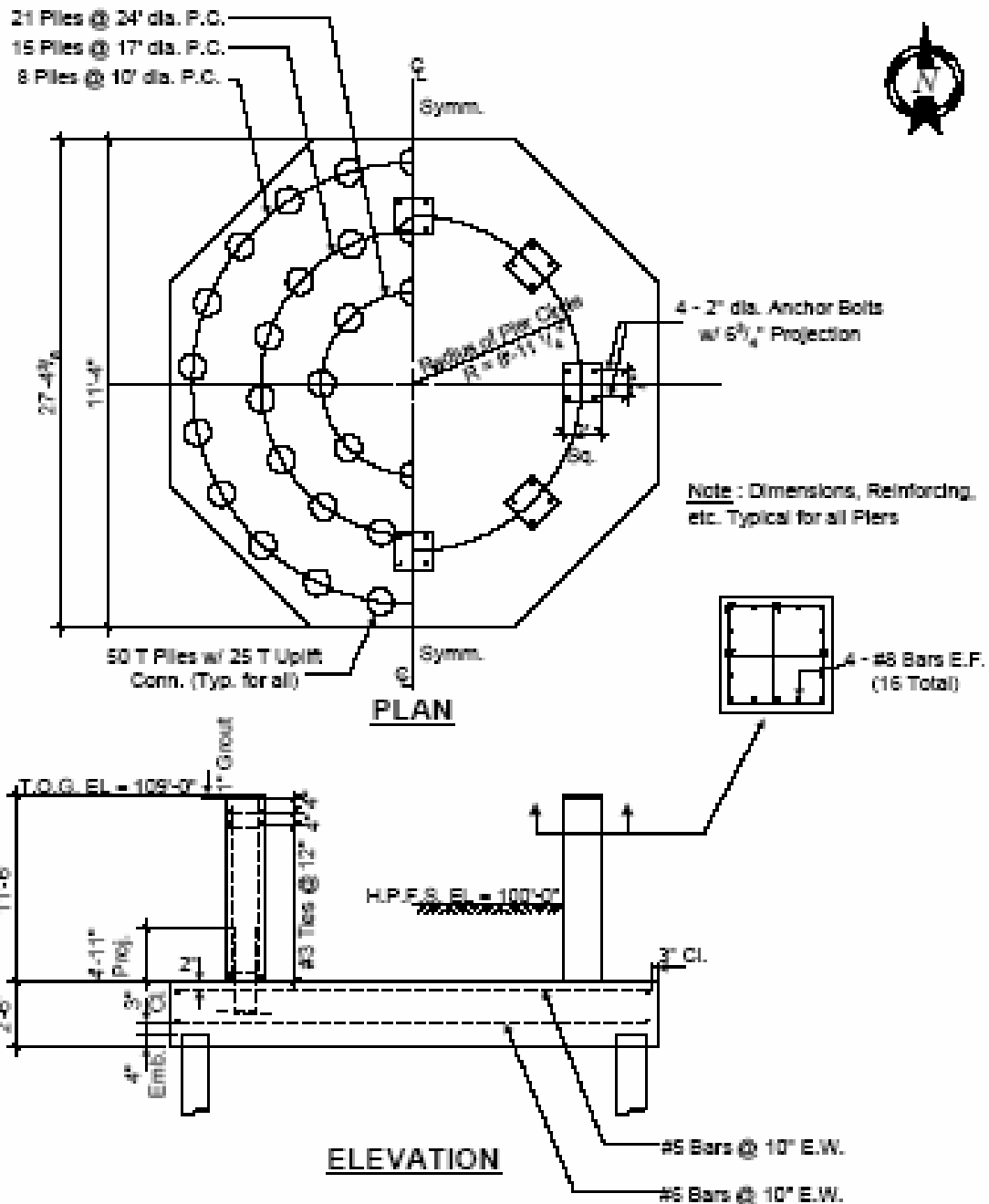
Use #8 Bars @ 12", Splice Length = 5' - 0" ($A_s = 0.79 \text{ in}^2 / \text{ft}$)

Top Longitudinal Reinforcing

Use #7 Bars @ 12", Splice Length = 3' - 9" (By Inspection)



SAMPLE DESIGN 7: CYLINDRICAL HEATER FOUNDATION ON PILES





SAMPLE DESIGN 7: CYLINDRICAL HEATER FOUNDATION ON PILES

DESIGN CONDITIONS

$$f'_c = 3,000 \text{ psi}$$

$$F_y = 60,000 \text{ psi}$$

$$\gamma_{\text{soil}} = 110 \text{ pcf}$$

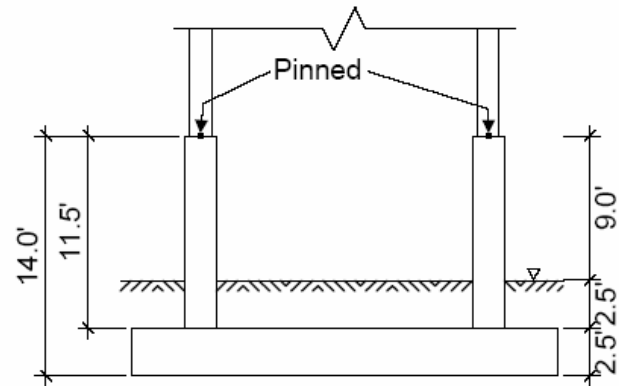
$$\gamma_{\text{conc}} = 150 \text{ pcf}$$

$$\text{Pile Capacity} = 50 \text{ T}$$

$$\text{Uplift Capacity} = 25 \text{ T}$$

$$\text{Lateral Capacity} = 2.0 \text{ k / Pile}$$

Passive Soil Resistance = 110 psf @ 1 ft
increasing 110 psf / ft depth



Note : Neglect wind on concrete columns.

VENDOR DATA

8 columns on a $17' - 10 \frac{1}{2}''$ dia. Circle

$$\text{Dead Load} = 8 \text{ Col}(46.16 \text{ k / Col}) = 369.3 \text{ k}$$

$$\text{Oper. Load} = 8 \text{ Col}(52.48 \text{ k / Col}) = 419.8 \text{ k}$$

$$\text{Wind Shear} = 8 \text{ Col}(12.5 \text{ k / Col}) = 100 \text{ k}$$

$$\text{Max Vertical Load due to Wind} = \pm 167.25 \text{ k / Col @ Top of Col}$$

$$\pm 167.25 = \frac{M_c}{l} \rightarrow I = \frac{Nd^2}{8} = \frac{8(17.875)^2}{8} = 319.5 \text{ ft}^2$$

$$M_{\text{wind}} = 167.25 \left(\frac{I}{c} \right) = 167.25 \left(\frac{319.5}{8.94} \right) = 5,977 \text{ ft-k @ Bottom of Heater Legs}$$

Note : Engineer should make a check of wind on heater to verify vendor loading.

2" dia. Anchor Bolts (4 per Col) @ 7" Spacing E.W.

Maximum Column Size per Vendor = 2' - 0" Square

PIER DESIGN

Note : For brevity, column design including column dimensions, anchor bolt design, and reinforcement are not shown. See Guideline 000-215-1212 "Concrete Column Design" for details on column design.

Use 4 - #8 Bars E.F. (16 Total) with #3 Ties @ 12" c/c



SAMPLE DESIGN 7: CYLINDRICAL HEATER FOUNDATION ON PILES

PILE CAP DESIGN

Minimum Pile Cap Thickness to Develop #8 Bars

$$l_{dh} = \frac{1,200d_b}{\sqrt{f'_c}} = \frac{1,200(1.00)}{\sqrt{3,000}} = 22"$$

$$T_{req'd} = 4" \langle Emb \rangle + 3" \langle Clr \rangle + 2(0.75") + 22" = 30" \leq T_{used}$$

$$\begin{aligned} \text{Minimum Octagon Size} &= \text{Pier Circle} + \text{Pier Width} + \text{Edge Distance} \\ &= 17.875 + 2.0 + 2(0.5) = 20.875' \end{aligned}$$

$$\text{Try } 27' - 4\frac{3}{8}" \text{ Octagon where } D = 27.361'$$

$$\begin{aligned} A &= 620.18 \text{ ft}^2 \\ \text{Assume } 2' - 6" \text{ Thick} \end{aligned}$$

$$\begin{aligned} \text{Wt. Columns} &= 8(2.0)^2(11.5)(0.15) = 55.20 \text{ k} \\ \text{Wt. Pile Cap} &= 620.18(2.5)(0.15) = 232.57 \text{ k} \\ \text{Wt. Soil} &= [620.18 - 8(2)^2](2.5)(0.11) = 161.75 \text{ k} \\ \text{Total} &= \underline{449.52 \text{ k}} \end{aligned}$$

$$\text{Buoyant Wt.} = 449.52 - 0.624[620.18(5 \text{ ft})] = 256.02 \text{ k}$$

Piles

Try 44 - 50 T Piles with Minimum c/c Pile Spacing = 3.5'

<u>Pile Circle</u>	<u>No. of Piles</u>	<u>I = (1/8)Nd²</u>
d3 = 24'	21	1,512.0
d2 = 17'	15	541.9
d1 = 10'	8	100.0
Total =	44	2,153.9 ft ²

Wind Shear = 100 k (Refer to Pg.2)

$$M_{wind} = 5,977 + 100(14') = 7,377 \text{ ft-k}$$

$$P = \frac{V}{N} \pm \frac{M_c}{I}$$

Oper. + Wind (Max Compression)

$$V = 419.8 \text{ k} + 449.52 \text{ k} = 869.32 \text{ k}$$



SAMPLE DESIGN 7: CYLINDRICAL HEATER FOUNDATION ON PILES

$$P_{\max} = \frac{869.32}{44} + \frac{7,377(12)}{2,153.9} = 60.9 \text{ k} < 100 \text{ k Allowable (okay)}$$

DL + Wind (Buoyant Condition)

$$V = 369.3 \text{ k} + 256.02 = 625.32 \text{ k}$$

$$P_{\min} = \frac{625.32}{44} - \frac{7,377(12)}{2,153.9} = \left. \begin{array}{l} -26.9 \text{ k @ } 12' \\ -14.9 \text{ k @ } 8.5' \\ -2.9 \text{ k @ } 5' \end{array} \right\} < 50 \text{ k Allowable (okay)}$$

LATERAL RESISTANCE

$$V_{\text{wind}} = 100 \text{ k}$$

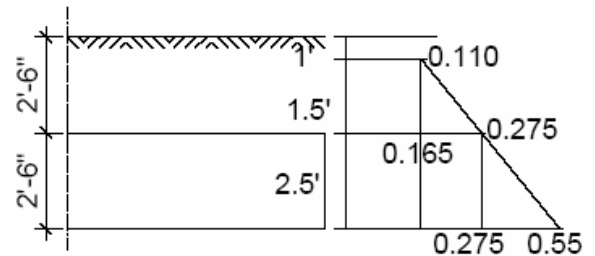
Passive Soil Pressure

$$R_p = 8(2')(1.5')(0.11 + \frac{0.165}{2}) + 27.36(2.5)(0.275 + \frac{0.275}{2}) = 32.8 \text{ K}$$

$$\text{Piles} = 44(2.0 \text{ k}) = 88.0 \text{ k}$$

$$\text{Total} = 32.8 \text{ k} + 88.0 \text{ k} = 120.8 \text{ k} > 100 \text{ k (okay)}$$

Use 44 - 50 T Piles with Uplift Connectors

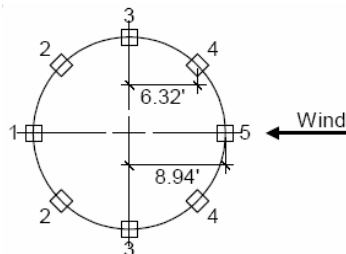


PILE CAP REINFORCEMENT

Critical Case → 0.9DL + 1.3 WL

Load Distribution on Columns

$$d = 17.875'$$



$$I = \frac{1}{8} \times Nd^2 = \frac{1}{8} (8)(17.875)^2 = 319.5 \text{ ft}^2$$

$$\text{Net Wt. Column} = 8(2')^2[0.15(9') + (0.15 - 0.11)(25)] = 46.4 \text{ k}$$

$$M_u = 1.3(5,977) = 7,770.1 \text{ ft-k}$$

SAMPLE DESIGN 7: CYLINDRICAL HEATER FOUNDATION ON PILES

$$P_u = 0.9(369.3 + 46.4) = 374.1 \text{ k}$$

$$P_{col} = \frac{374.1}{8} \pm \frac{7,770.1(c)}{319.5}$$

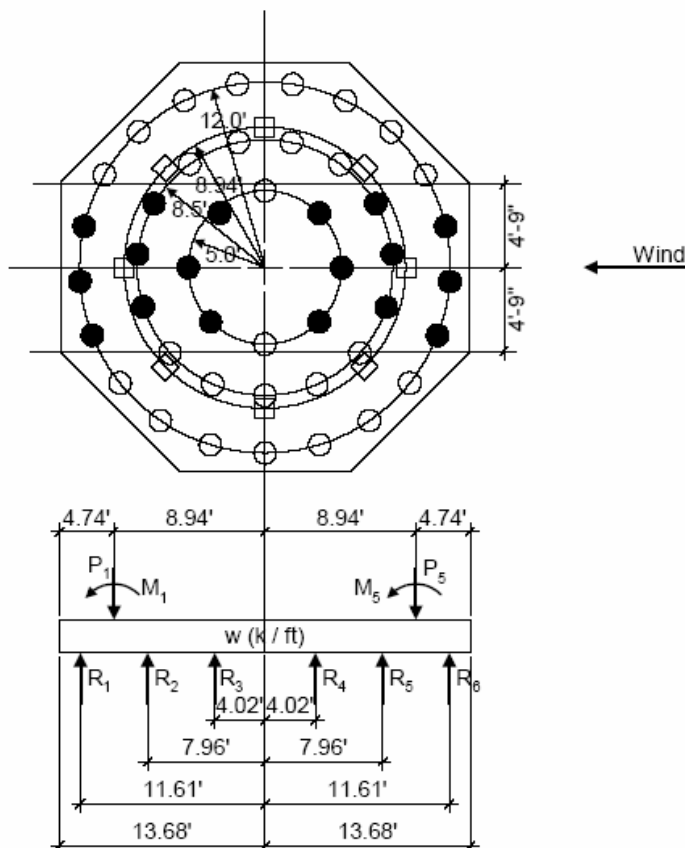
Column No.	C	P _{col}
1	+8.64	264.2
2	+6.32	200.5 x 2 = 401.0 k
3	0	46.8 x 2 = 93.6 k
4	-6.32	-106.9 x 2 = -213.8 k
5	-8.94	-170.7

$$M_{u,col} = 1.3(12.5 \text{ k})(14') = 227.5 \text{ ft-k}$$

$$\text{Ult. Wt. of Soil + Pile Cap, } w_u = 0.9[2.5(0.15) + 2.5(0.11)] = 0.585 \text{ ksf}$$

Use Strip Analysis as shown

$$\begin{aligned} \text{Width of Strip} &= 3 \times \text{Ftg.} + \text{Col. Wd.} \\ &= 3(2.5) + 2 \\ &= \underline{9' - 6''} \end{aligned}$$



Moment of Inertia of Pile Group

$$\begin{aligned} I &= 6(11.61)^2 + 6(7.96)^2 + 6(4.02)^2 \\ &= 1286 \text{ ft}^2 \end{aligned}$$



SAMPLE DESIGN 7: CYLINDRICAL HEATER FOUNDATION ON PILES

Dead Load + Wind (Refer to Pg.7)

$$w = 9.5'(0.585 \text{ ksf}) = 5.56 \text{ k / ft}$$

$$M_{\text{col}} = 227.5 \text{ ft-k}$$

$$P_1 = 264.2 \text{ k}$$

$$P_5 = -170.7 \text{ k}$$

$$\text{Total Load} = 5.56 \text{ k / ft}(27.361) + 264.2 - 170.7 = 245.6 \text{ k}$$

$$\text{Total Moment} = (227.5 \times 2) + 264.2(8.94) + 170.7(8.94) = 4,343.0 \text{ ft-k}$$

$$R = \frac{245.6}{18} \pm \frac{4,343.0(c)}{1,286}$$

$$R_1 = 13.64 + \frac{4,343.0(11.61)}{1,286} = 52.85 \text{ k} \times 3 = 158.5 \text{ k}$$

$$R_2 = 13.64 + \frac{4,343.0(7.96)}{1,286} = 40.52 \text{ k} \times 3 = 121.6 \text{ k}$$

$$R_3 = 13.64 + \frac{4,343.0(4.02)}{1,286} = 27.22 \text{ k} \times 3 = 81.7 \text{ k}$$

$$R_4 = 13.64 - \frac{4,343.0(4.02)}{1,286} = 0.06 \text{ k} \times 3 = 0.2 \text{ k}$$

$$R_5 = 13.64 - \frac{4,343.0(7.96)}{1,286} = -13.24 \text{ k} \times 3 = -39.7 \text{ k}$$

$$R_6 = 13.64 - \frac{4,343.0(11.61)}{1,286} = -25.57 \text{ k} \times 3 = -76.7 \text{ k}$$

Bottom Steel

$$M_u = \frac{360.73 \text{ ft-k}}{9.5'} = 38.0 \text{ ft-k / ft}$$

$$d = 30" - 4" < \text{Emb.} > - 3" < \text{Clr.} > - 1.5" = 21.5"$$

$$F = \frac{bd^2}{12,000} = \frac{12(21.5)^2}{12,000} = 0.46$$

$$K_u = \frac{M_u}{F} = \frac{38.0}{0.46} = 82.6 \rightarrow a_u = 4.42$$

$$A_s = \frac{M_u}{a_u \times d} = \frac{38.0}{4.42 \times 21.5} = 0.40 \text{ in.}^2 / \text{ft}$$

$$\frac{4}{3} A_s = 0.53 \text{ in.}^2 / \text{ft} \leftarrow \text{Governs}$$



SAMPLE DESIGN 7: CYLINDRICAL HEATER FOUNDATION ON PILES

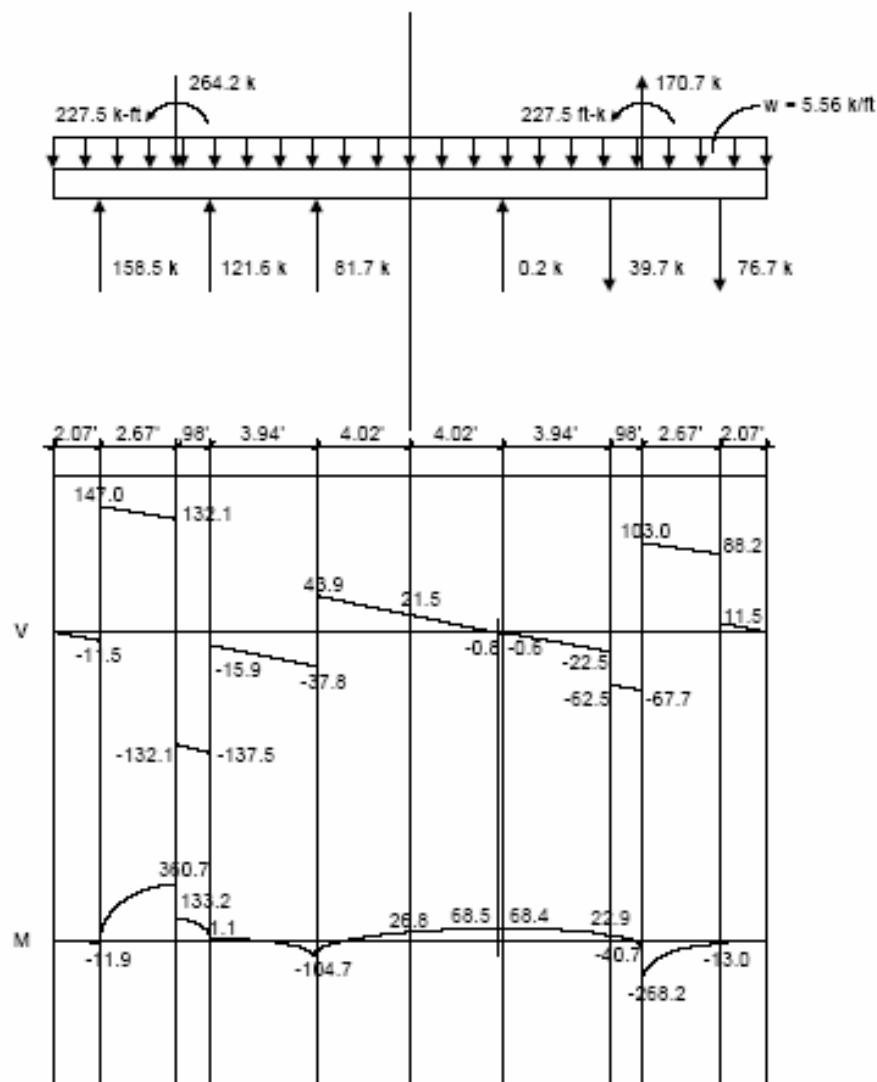
$$A_{s,min} = 0.0033(12)(21.5) = 0.85 \text{ in.}^2 / \text{ft}$$

Use #6 Bars @ 10" E.W. ($A_s = 0.53 \text{ in.}^2$)

Maximum Shear = 147.0 k

Maximum Positive Moment = 360.7 ft-k

Maximum Negative Moment = -268.2 ft-k





SAMPLE DESIGN 7: CYLINDRICAL HEATER FOUNDATION ON PILES

Top Steel

$$M_u = \frac{268 ft - k}{9.5'} = 28.2 \text{ ft-k / ft}$$

$$d = 30" - 2" < \text{Clr.} > - 1.5" = 26.5"$$

$$F = \frac{bd^2}{12,000} = \frac{12(26.5)^2}{12,000} = 0.70$$

$$K_u = \frac{M_u}{F} = \frac{28.2}{0.70} = 40.3 \rightarrow a_u = 4.45$$

$$A_s = \frac{M_u}{a_u \times d} = \frac{28.2}{4.45 \times 26.5} = 0.24 \text{ in.}^2 / \text{ft}$$

$$\frac{4}{3} A_s = 0.32 \text{ in.}^2 / \text{ft} \leftarrow \text{Governs}$$

$$A_{s,\min} = 0.0033(12)(26.5) = 1.05 \text{ in.}^2 / \text{ft}$$

Use #5 Bars @ 10" E.W. ($A_s = 0.37 \text{ in.}^2$)

Shear

$$V_{u,\max} = \frac{147.0k}{9.5} = 15.5 \text{ k / ft}$$

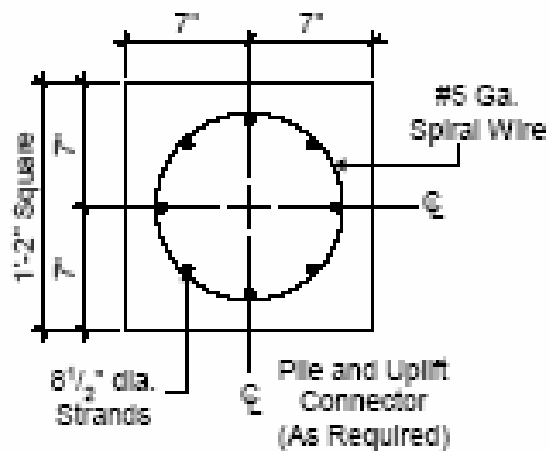
$$d = 21.5"$$

$$v_u = \frac{15.5 \times 1000}{0.85(12)(21.5)} = 70.7 \text{ psi} < 2\sqrt{f'_c} = 109.5 \text{ psi (okay)}$$

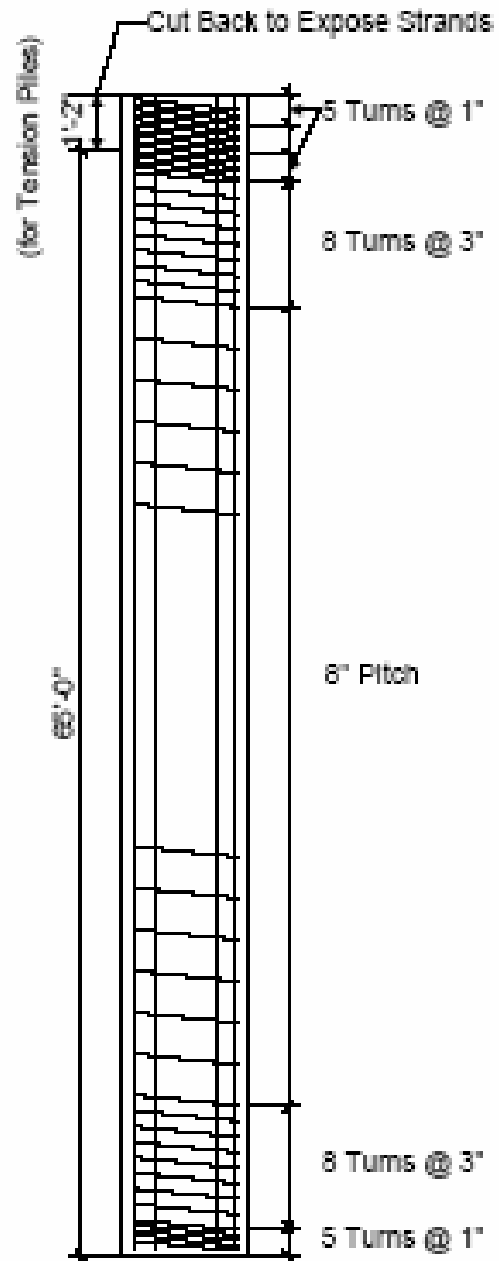
Punching Shear (okay by inspection)



SAMPLE DESIGN 8 : STRUCTURAL DESIGN OF A PRESTRESSED CONCRETE PILE



PLAN



ELEVATION



SAMPLE DESIGN 8 : STRUCTURAL DESIGN OF A PRESTRESSED CONCRETE PILE

DESIGN DATA

References

ACI 318-89
UBC-88
Project Soil Report

Materials

Piles : 14" Square Prestressed Concrete Piles x 65' - 0" Long
(with 1' - 2" Head Extension for Tension Connector)

Capacities : Compression = 50T
 Tension = 23 T
 Lateral = 2 T (Free Head Condition; Max Moment = 24 ft-k @ 6" Depth)

Concrete : $f'_c = 5,000$ psi @ 28 days
 = 4,000 psi @ Release
 $\gamma_{conc} = 150$ pcf

Steel : Rebar - ASTM A-615 Grade 60
 Wire - No. 5 Ga. Minimum

 Prestressed Strands - $\frac{1}{2}$ " dia., $F_y = 270$ ksi

Soil : Uniform Medium Clay from Grade to EL 70' - 0"

PILE DESIGN

Check Driving Stress

Normally, the pile is checked by the soil consultant for driving stress using the Wave Equation. See Bowles, Foundation Analysis and Design for procedure on the Wave Equation. In general, piles are driven to their compressive capacity multiplied by a factor of safety.

Use Factor of Safety = 2.0

$$P_u = 2 \times (50 \times 2) = 200 \text{ k}$$

$$A = (14)^2 = 196 \text{ in.}^2$$

$$f_c = \frac{P_u}{A} = \frac{200}{196} = 1.02 \text{ ksi}$$

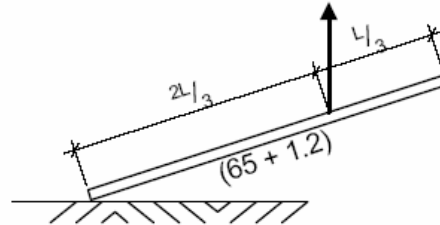
$$\frac{f_c}{f'_c} = \frac{1,020}{5,000} = 0.204 < 0.6 \leftarrow \text{Allowable Dynamic Stress Ratio per Bowles, No Code Requirement}$$

SAMPLE DESIGN 8 : STRUCTURAL DESIGN OF A PRESTRESSED CONCRETE PILE**Check Handling Stress**

$$w = (1.17)^2 \times 0.15 = 0.205 \text{ klf}$$

$$M_{\max} = w \times \frac{L^2}{18} = 0.205 \times \frac{(66.2)^2}{18} = 49.9 \text{ ft-k L}$$

$$S = \frac{1}{6} \times (14)^3 = 457 \text{ in.}^3$$



$$f_b = \frac{49.9(12)}{457} = 1.31 \text{ ksi} > f_{pc, \min} = 0.70 \text{ ksi} \leftarrow \text{per UBC; No Good for Tensile Stress}$$

Allowable Tensile Stress in Concrete per ACI-318 = $6 \sqrt{f'_c}$

$$f_{pc, req'd} = f_b - 6 \sqrt{f'_c} = 1.31 - \frac{6 \times \sqrt{5,000}}{1,000} = 0.886 \text{ ksi}$$

Determine the Number of Strands Required for Handling Stress

$$A_s = 0.153 \text{ in.}^2 / \text{strand}$$

$$f_{ps} = 0.7 f_{pu} = 0.7 \times 270 = 189 \text{ ksi}$$

Stress Losses = 30 ksi $\leftarrow$ Assumed per UBC

$$f_{se} = 189 - 30 = 159 \text{ ksi}$$

$$\text{No. of Strands Req'd} = \frac{A \times f_{pc, req'd}}{A_s \times f_{se}} = \frac{(14)^2 \times 0.886}{0.153 \times 159} = 8$$

$$\text{Use } 8 - 1 \frac{1}{2} \text{ " dia. 7 - Wire Strands (Avg. Prestress in Concrete } = f_{pc} = \frac{8 \times 0.153 \times 159}{196} = 1.00 \text{ ksi}$$

Check Compressive Stress

$$\sigma_c = \frac{P}{A} = \frac{100}{196} = 0.51 \text{ ksi}$$

$$\text{Per UBC : } f_c = 0.33 f'_c - 0.27 f_{pc} = 0.33(5.0) - 0.27(1.0) = 1.38 \text{ ksi} > 0.51 \text{ ksi (okay)}$$

Check Uplift Stresses

$$\sigma_c = \frac{T}{A} = \frac{-46}{196} = 0.235 \text{ ksi} < f_{pc} = 1.0 \text{ ksi (okay)}$$



SAMPLE DESIGN 8 : STRUCTURAL DESIGN OF A PRESTRESSED CONCRETE PILE

Check Compression and Bending

$$\sigma_c = 0.51 + \frac{24(12)}{457} = 1.14 \text{ ksi} < 1.38 \text{ ksi (okay)}$$

Check Uplift and Bending

$$\sigma_c = -0.235 - \frac{24(12)}{457} = -0.865 \text{ ksi} < f_{pc} \leftarrow \text{Section is in Compression (okay)}$$

REINFORCEMENT

Per UBC Sec.2909(e)

UPLIFT CONNECTORS DESIGN

$$\text{Uplift Capacity} = 23 T = 46 \text{ k}$$

$$T_u = 46 \times 1.7 = 78.2 \text{ k}$$

$$\begin{aligned} \text{No. of Strands to be Developed} \\ &= \frac{78.2}{270 \times 0.153} = 2 \text{ Strands} \end{aligned}$$

$$A_{c, \text{req'd}} = \frac{2 \times 270 \times 0.153 \times 1,000}{0.65 \times 4 \times \sqrt{3,000}} = 580 \text{ in.}^2$$

$$\begin{aligned} A_{c, \text{prov'd}} &= (2 \times 10 + 2.25) \left(10 + \frac{5.25}{2} + 4.5 \right)^2 \\ &= 762 \text{ in.}^2 > 580 \text{ in.}^2 \text{ (okay)} \end{aligned}$$

Use 2 - Uplift Connectors per Pile; Locate Connectors @ 180" on Strand Circle

